Recursive Computations for Khovanov-Rozansky Homology

Misha Mazin

**Partially based on previous work with Nicolle González, Matt Hogancamp, and Carmen Caprau.*

Categorified Young symmetrizers (Elias-Hogancamp)

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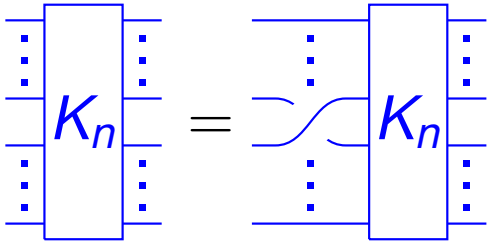
1. 

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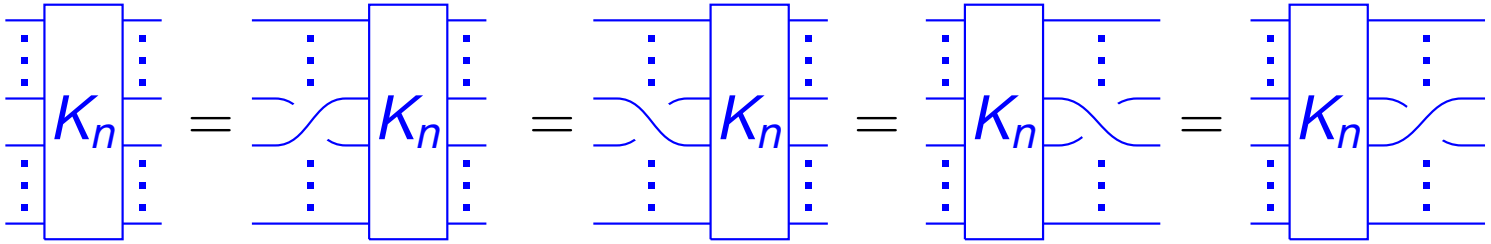
1. 

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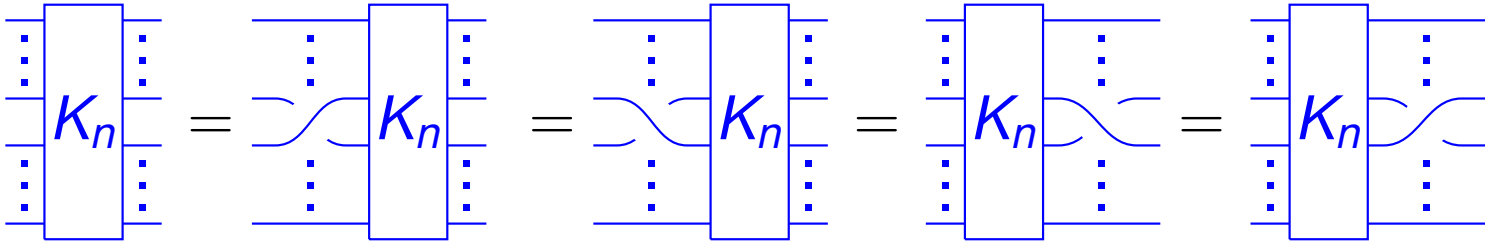
1. 

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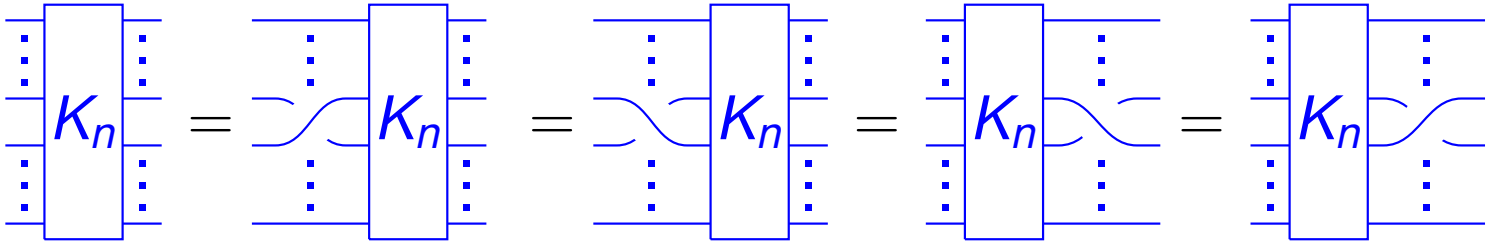
1. 

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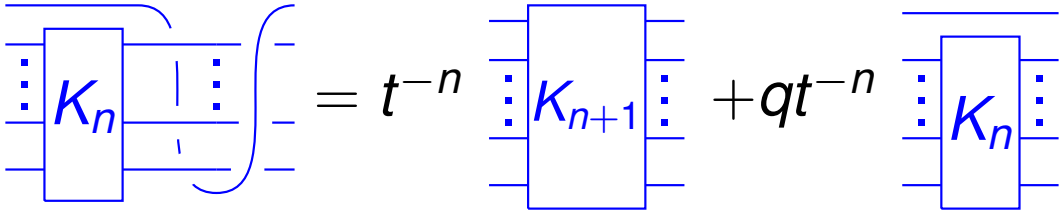

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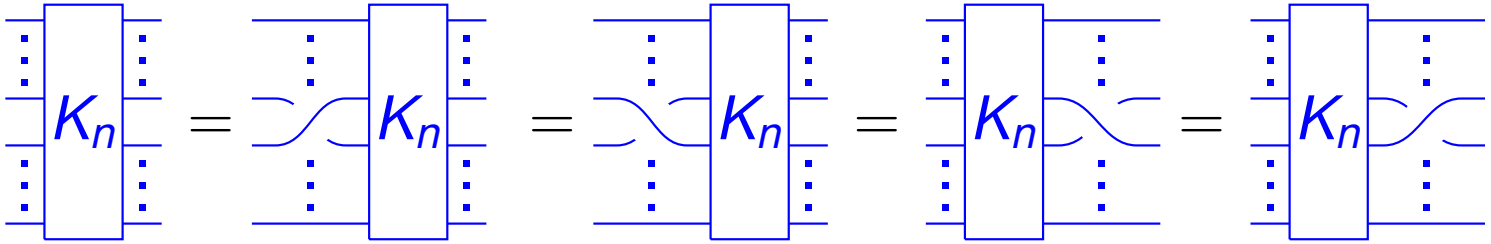
1. 

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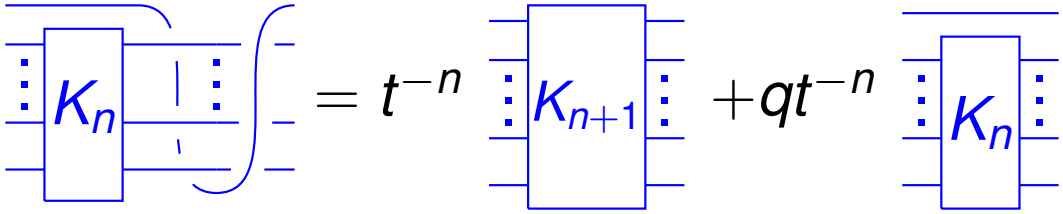

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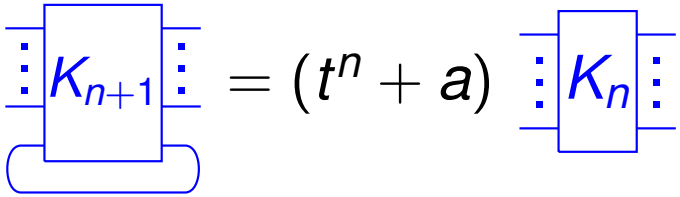
3. 

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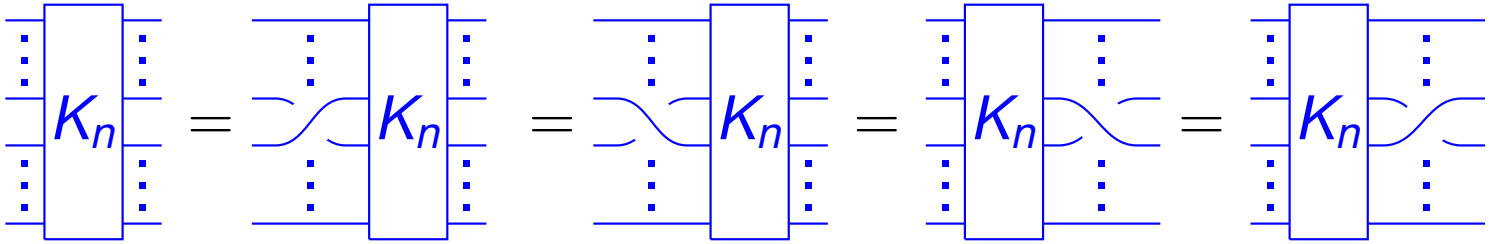
1. 

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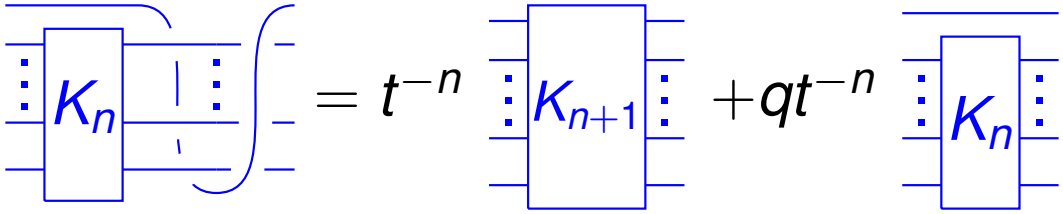

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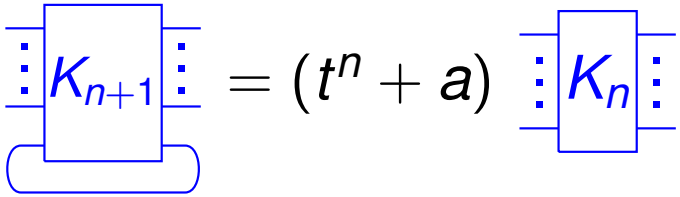
3. 

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Remark

In particular, from (2) with $n = 0$ one gets

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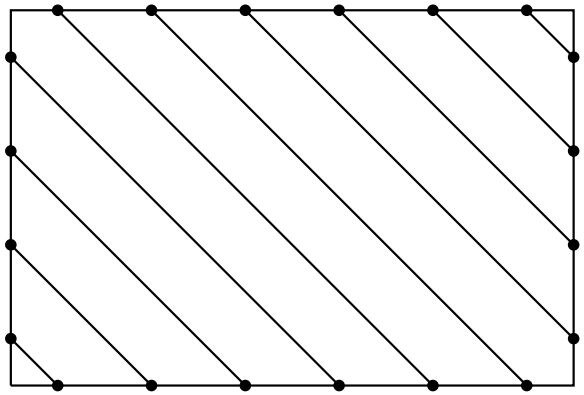
Hogancamp-Mellit recursions



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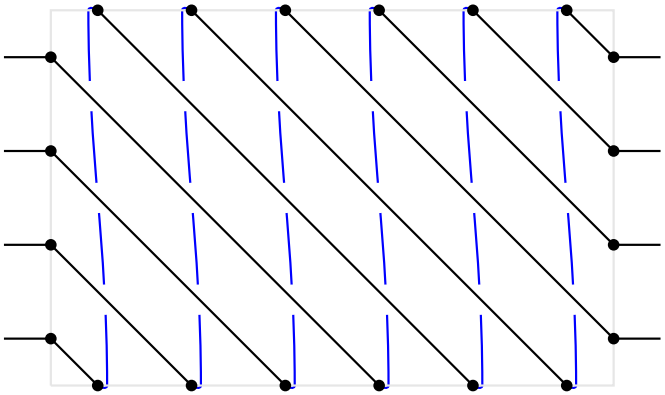

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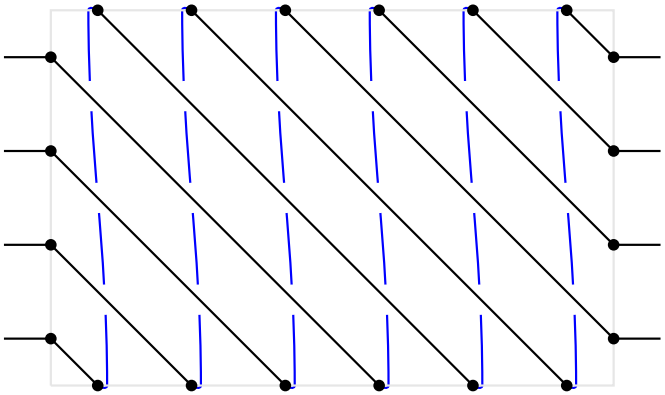

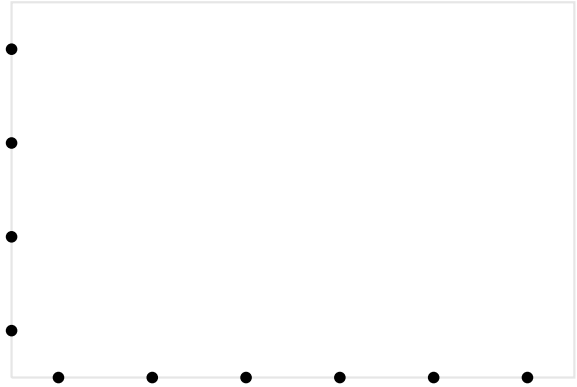

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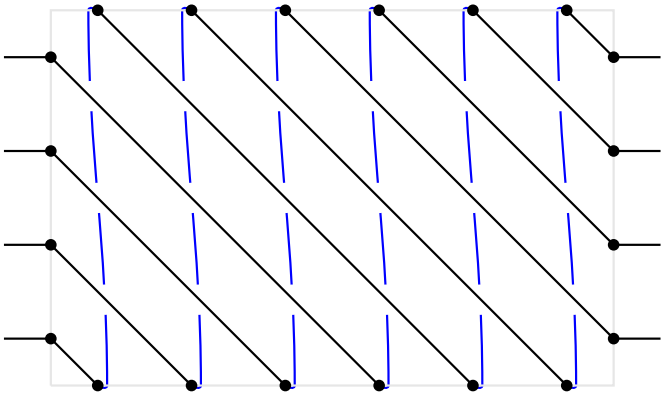

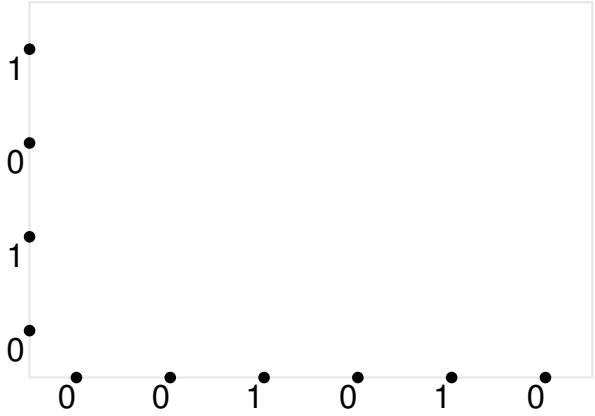

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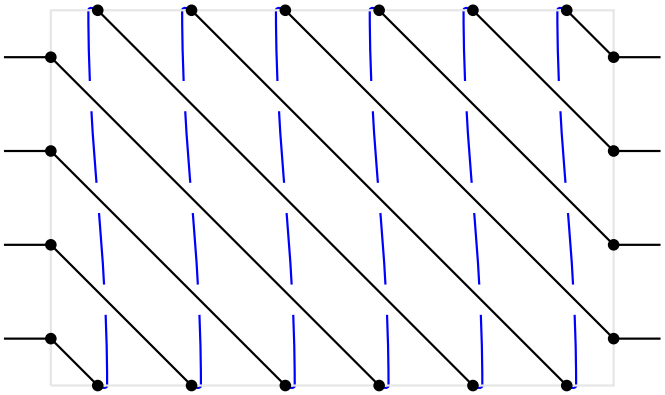

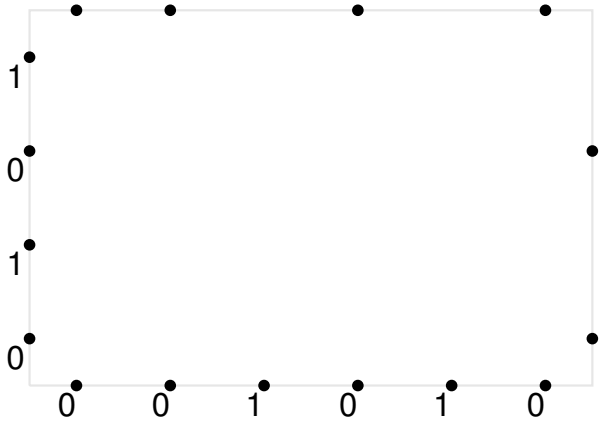

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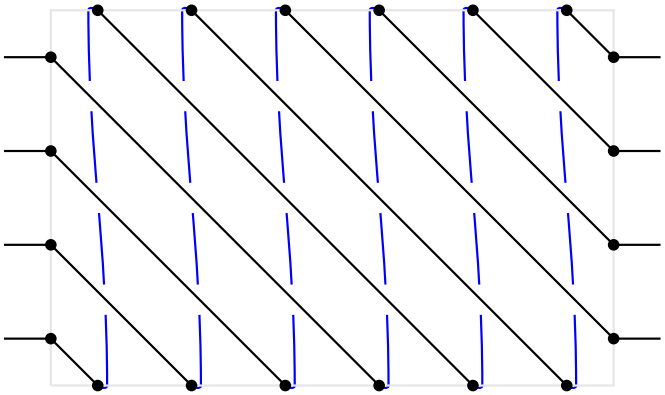

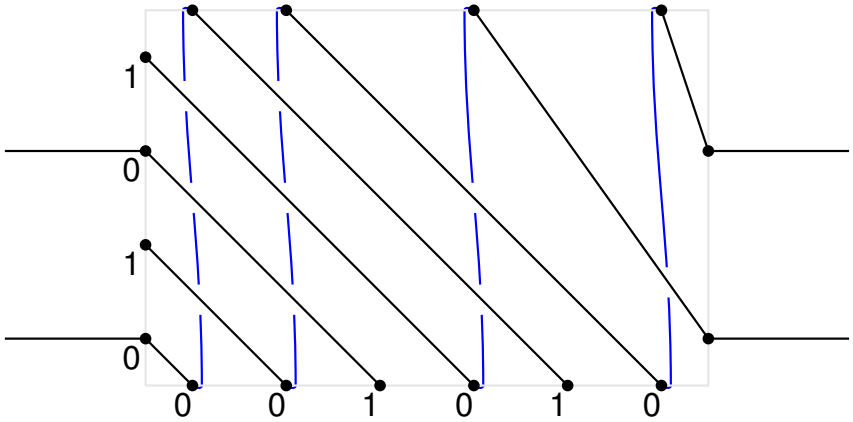

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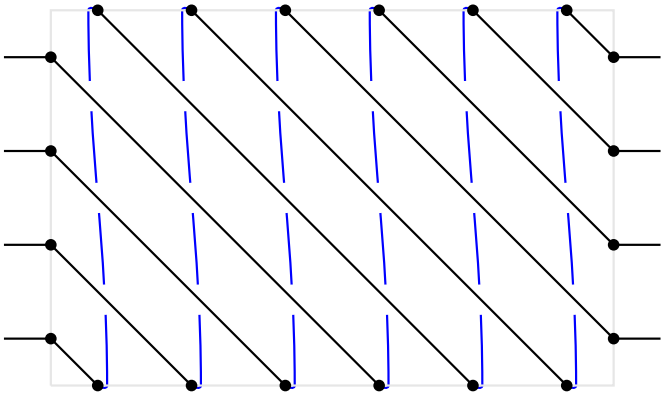

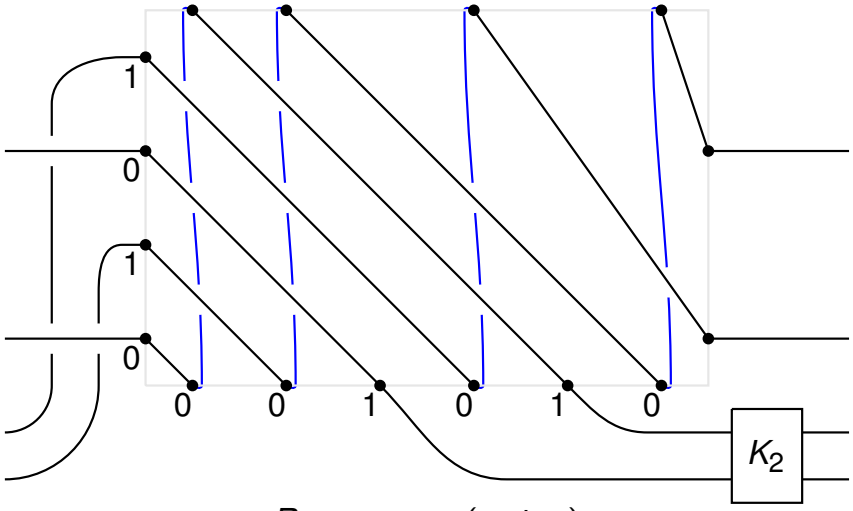

$R_{0101,001010}(q,t,a)$

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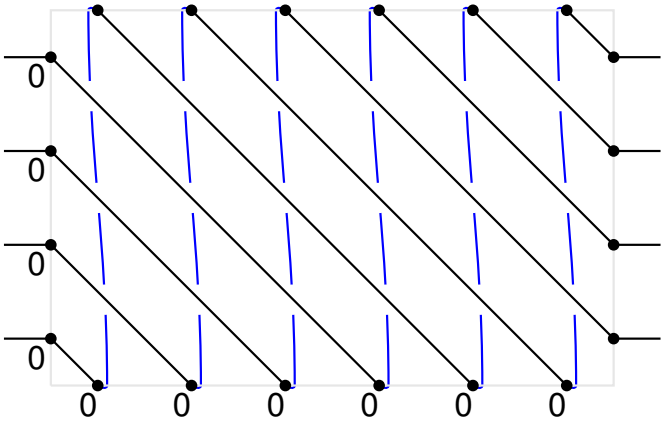
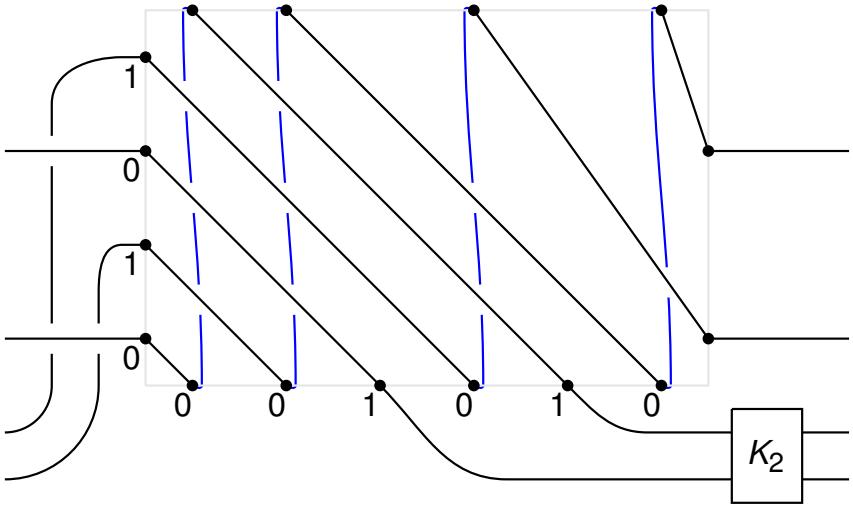


$R_{0000,000000}(q,t,a)$



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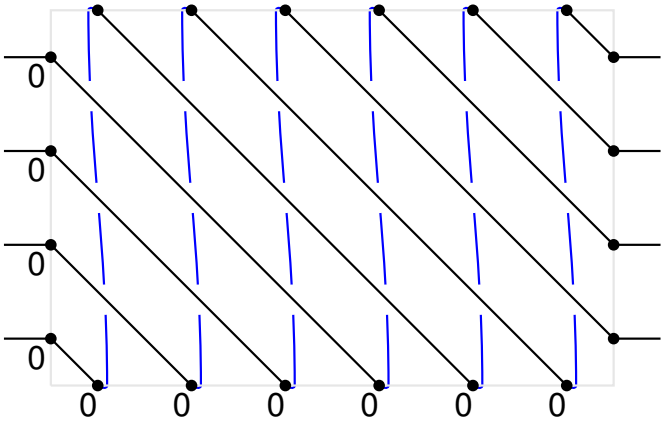

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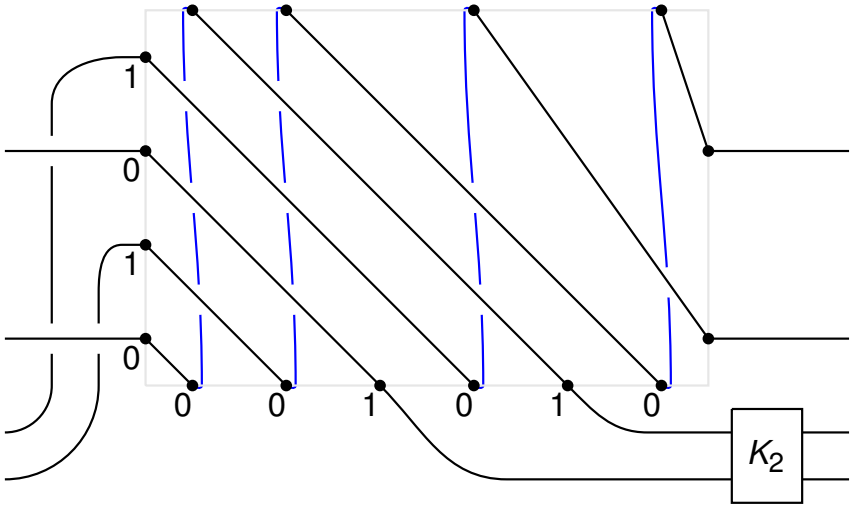


$R_{0101,001010}(q,t,a)$

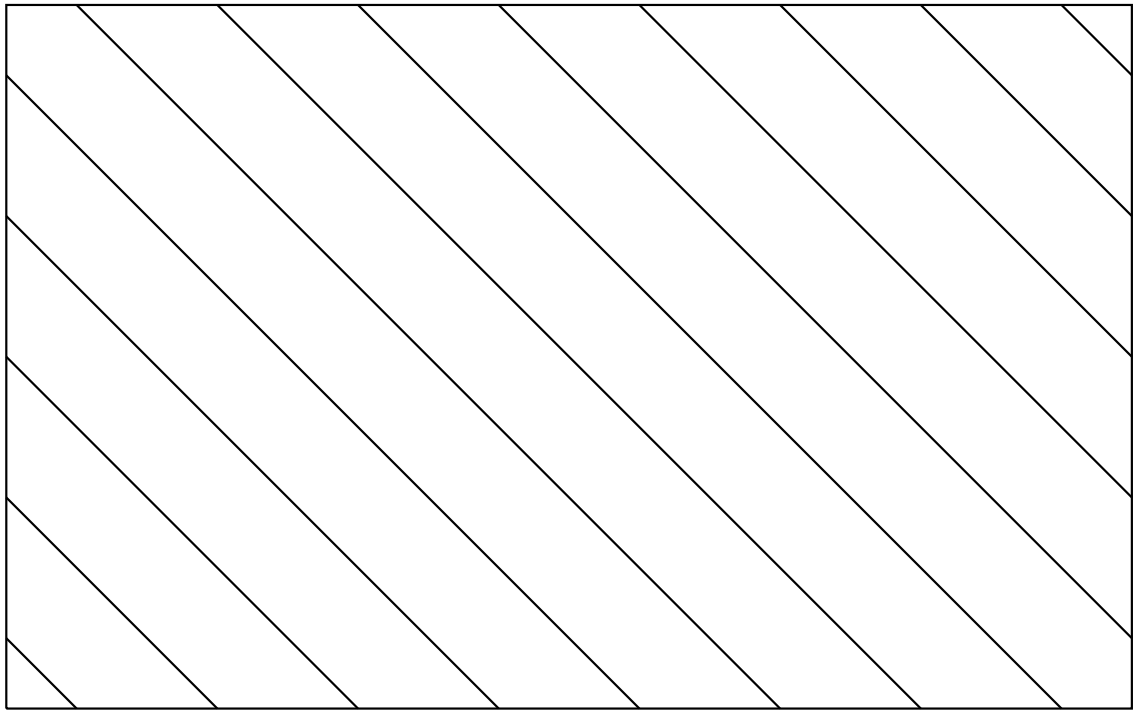
Theorem (Hogancamp-Mellit)

$$R_{0\mathbf{u},0\mathbf{v}} = t^{-|\mathbf{u}|}R_{\mathbf{u}1,\mathbf{v}1} + qt^{-|\mathbf{u}|}R_{\mathbf{u}0,\mathbf{v}0}, \quad R_{1\mathbf{u},0\mathbf{v}} = R_{\mathbf{u}1,\mathbf{v}}, \quad R_{\emptyset,0^n} = \left(\frac{1+a}{1-q}\right)^n,$$

$$R_{1\mathbf{u},1\mathbf{v}} = (t^{|\mathbf{u}|}+a)R_{\mathbf{u},\mathbf{v}}, \quad R_{0\mathbf{u},1\mathbf{v}} = R_{\mathbf{u},\mathbf{v}1}, \quad R_{0^m,\emptyset} = \left(\frac{1+a}{1-q}\right)^m,$$

where $R_{\emptyset,\emptyset} := 1$.

Shortcut torus knots



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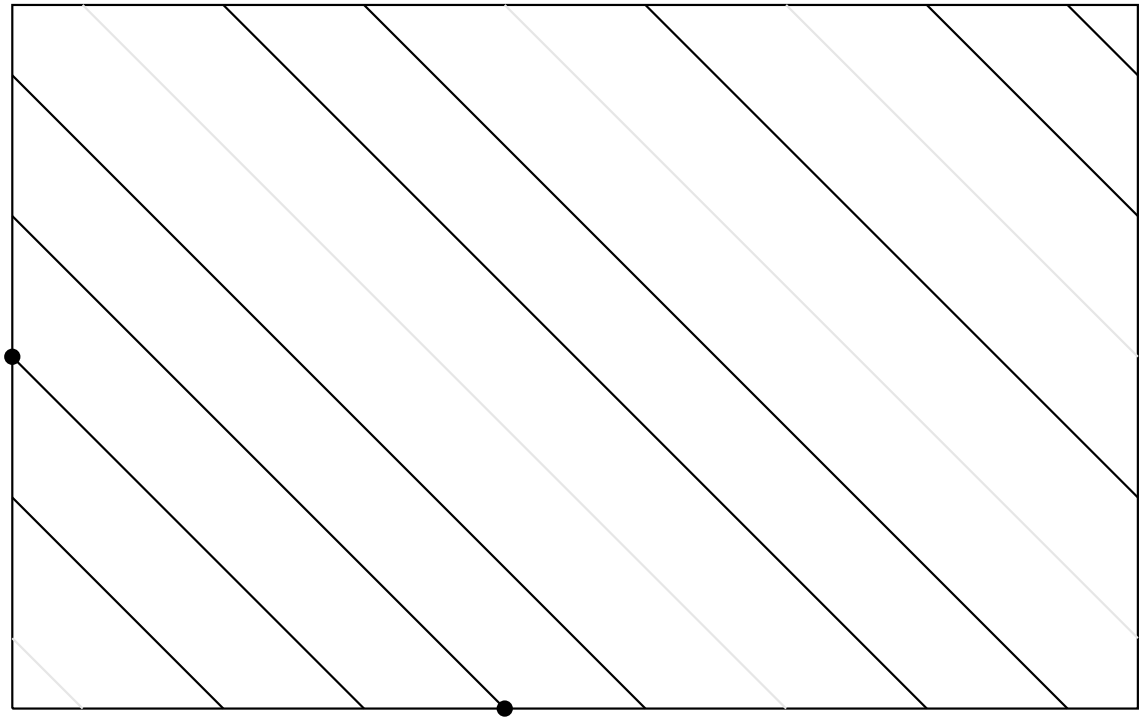

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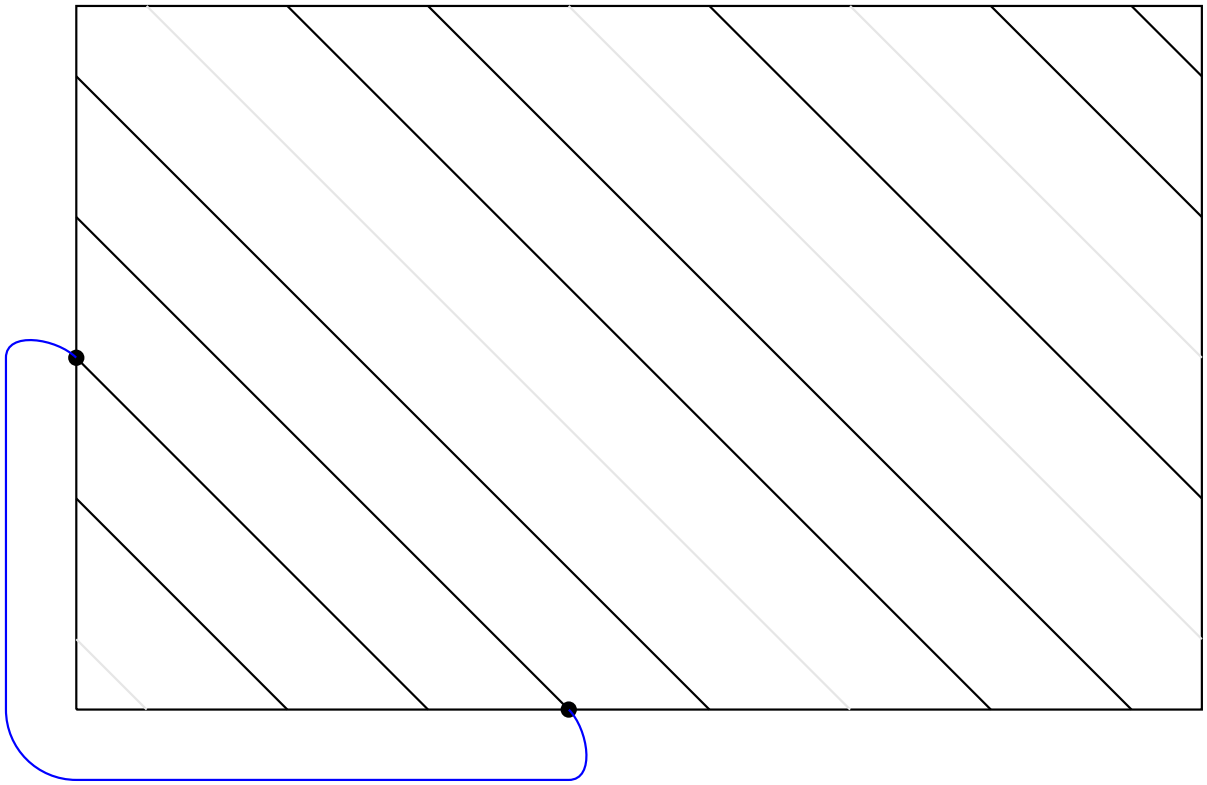

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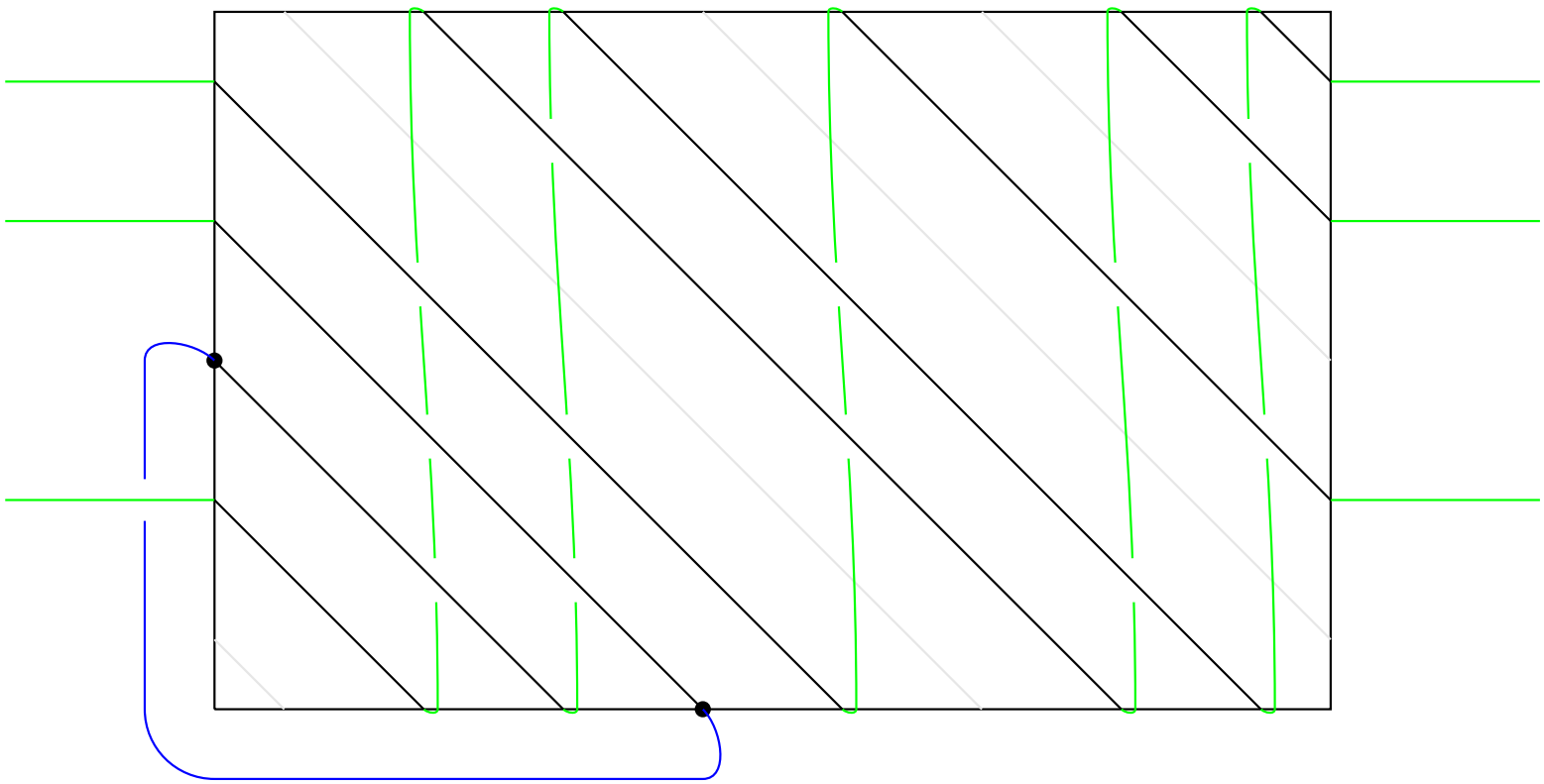

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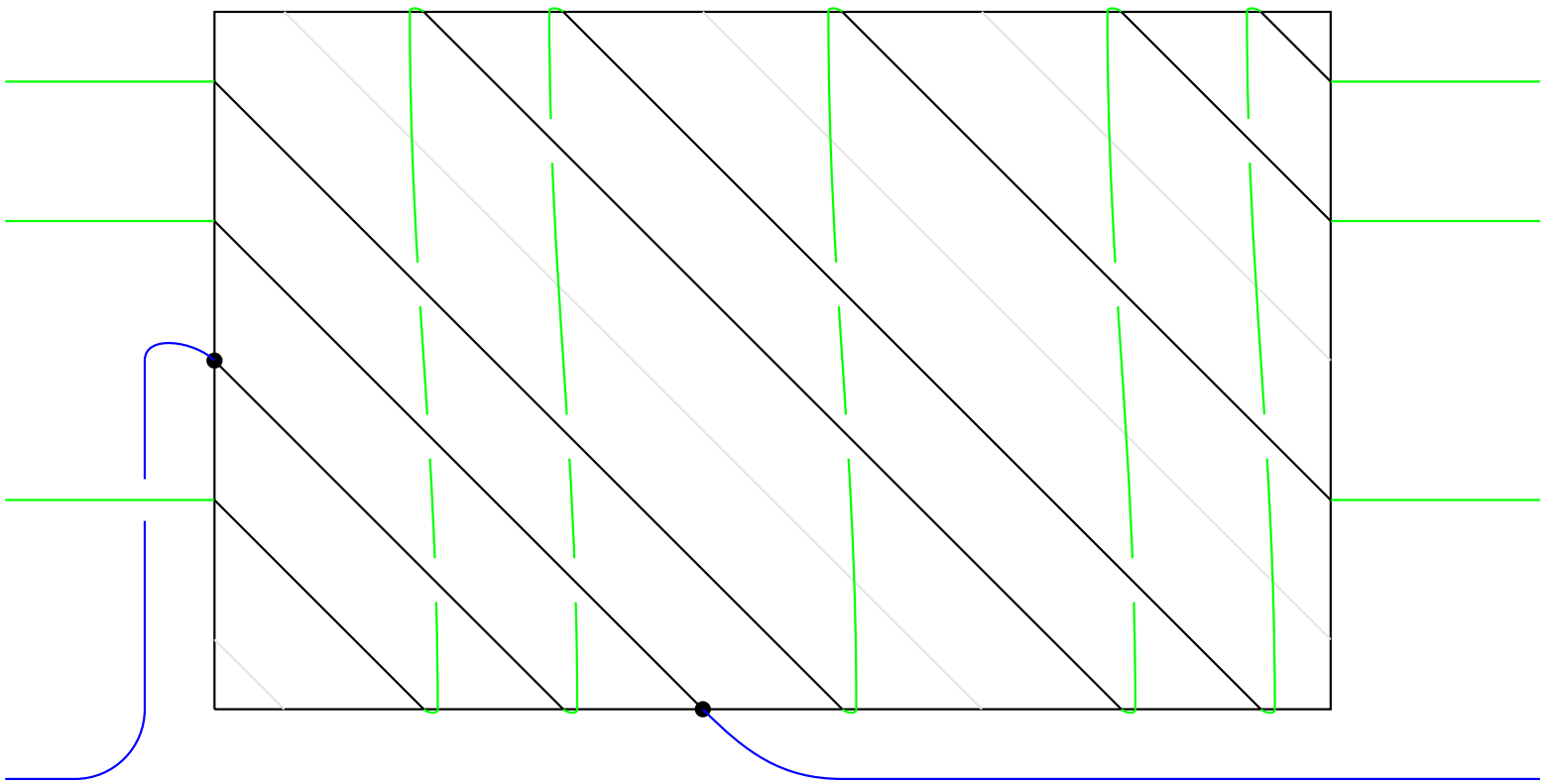

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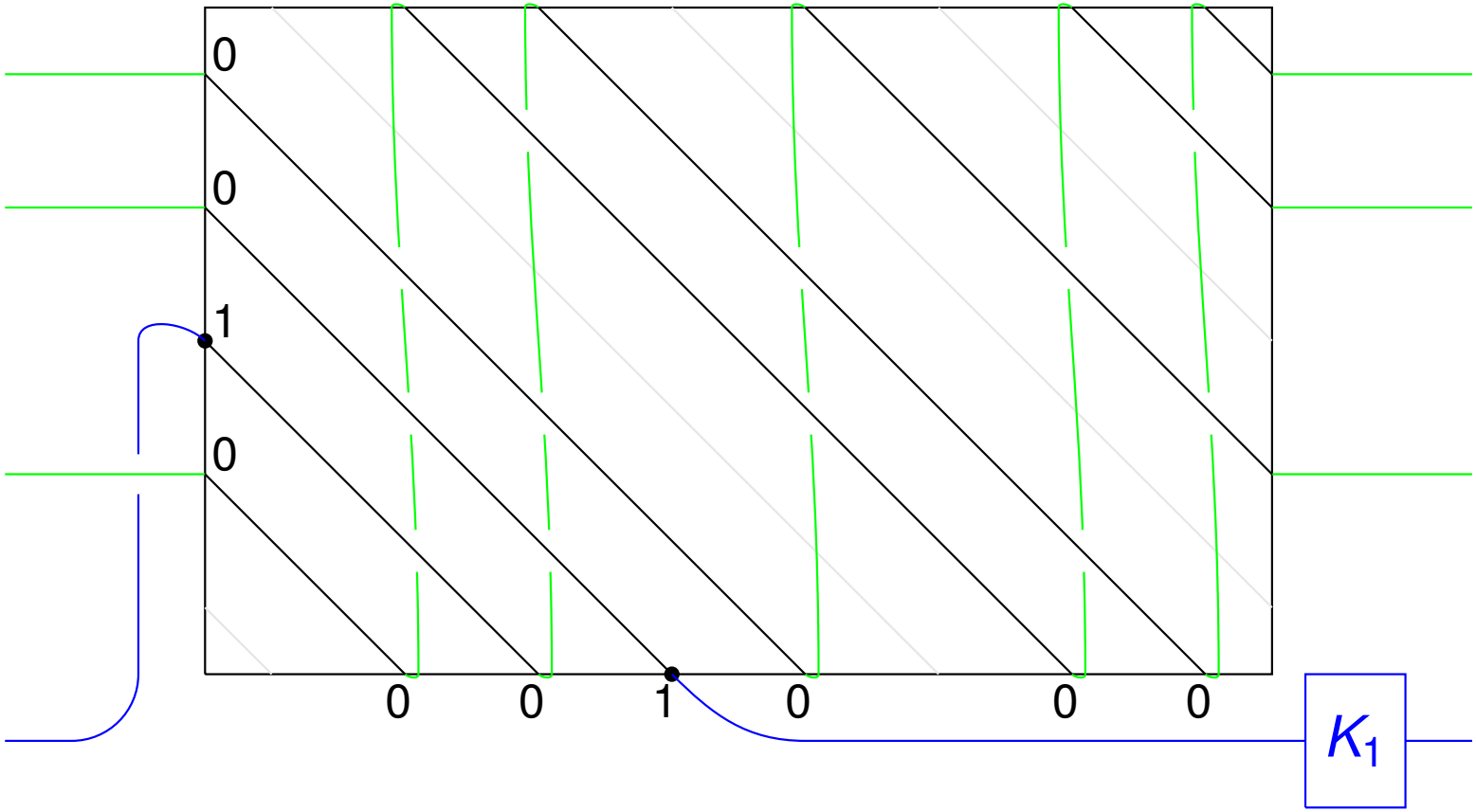

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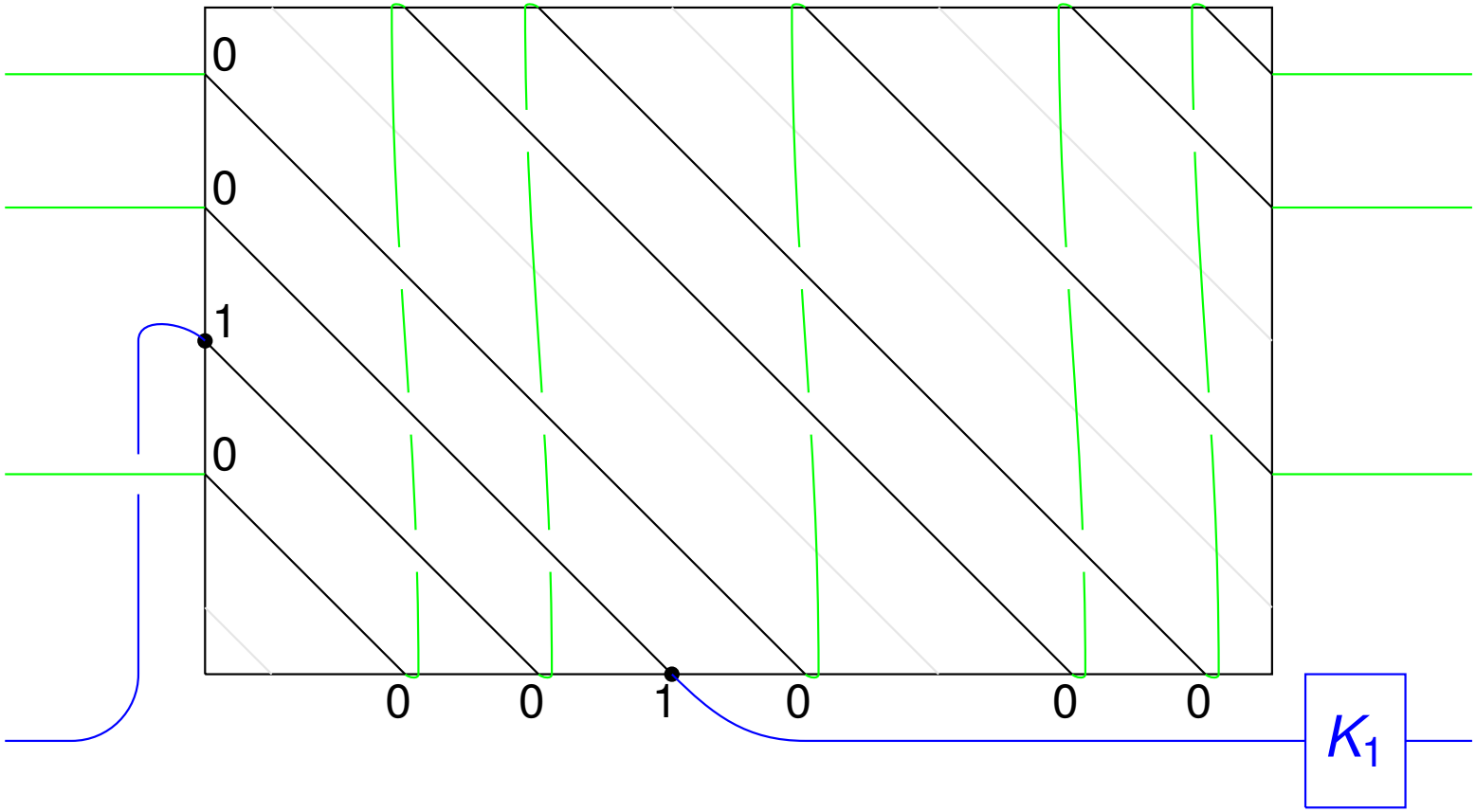
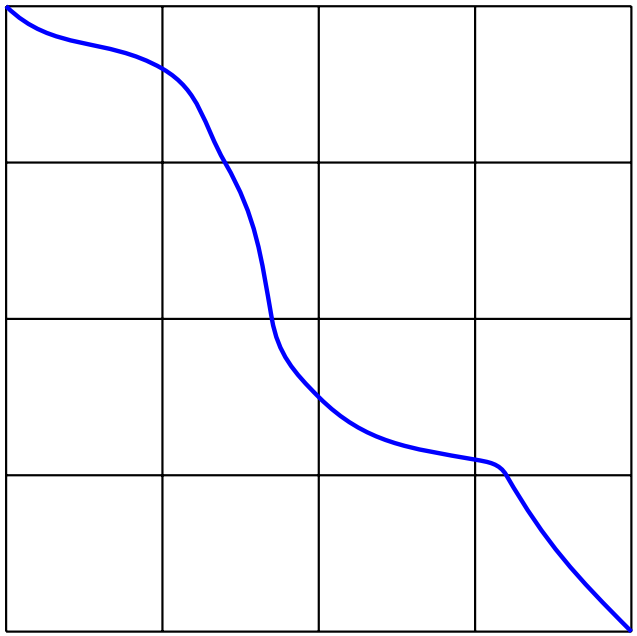


Theorem

All knots that appear in the Hogancamp-Mellit recursion are the shortcut torus knots (as above).

Monotone knots of Galashin-Lam



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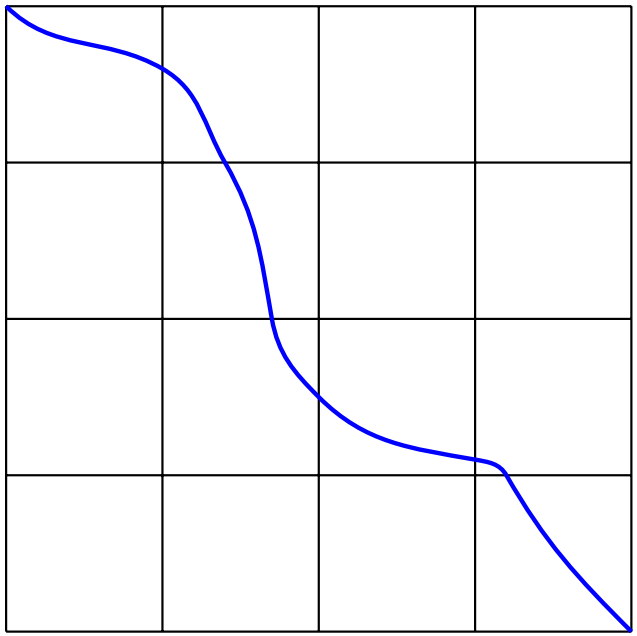

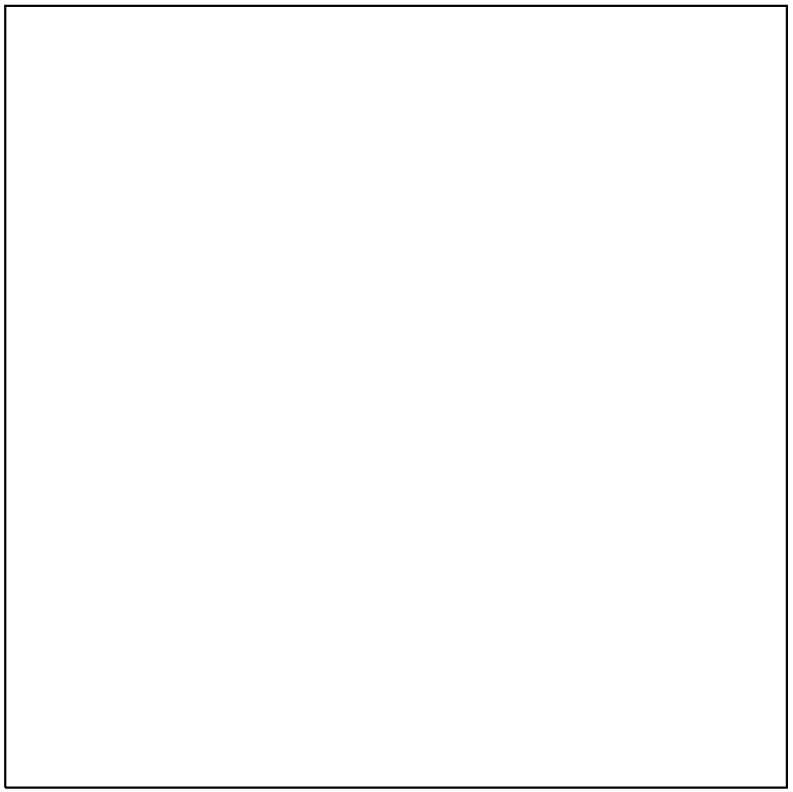

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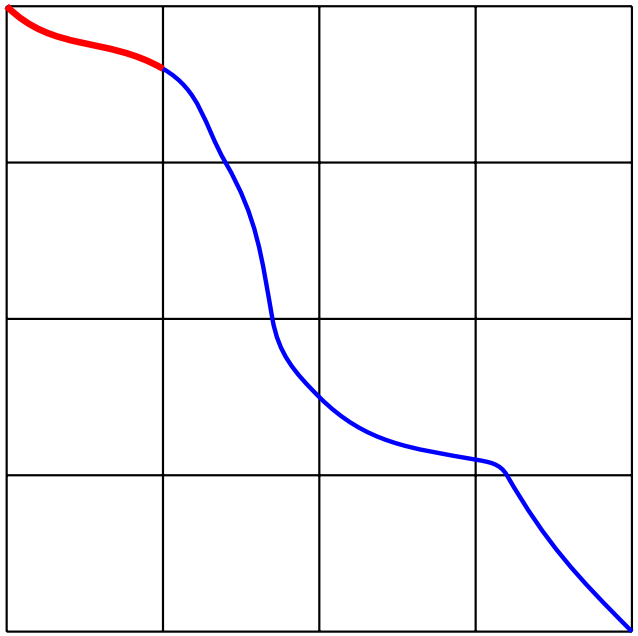

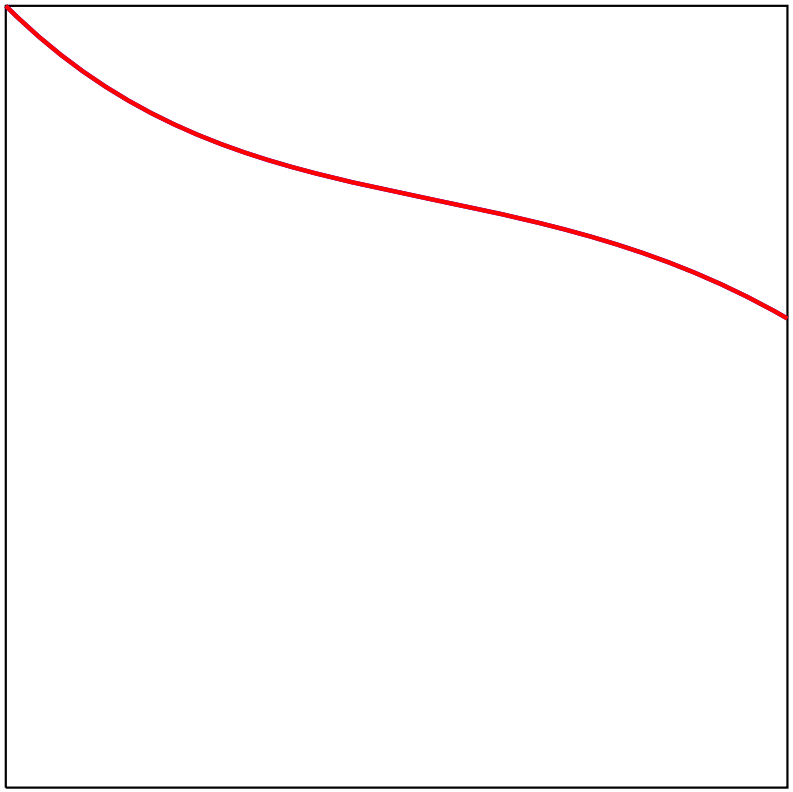

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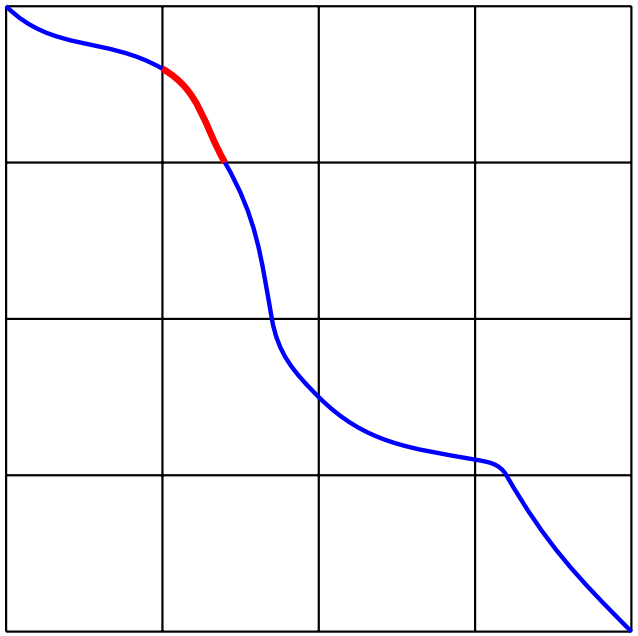

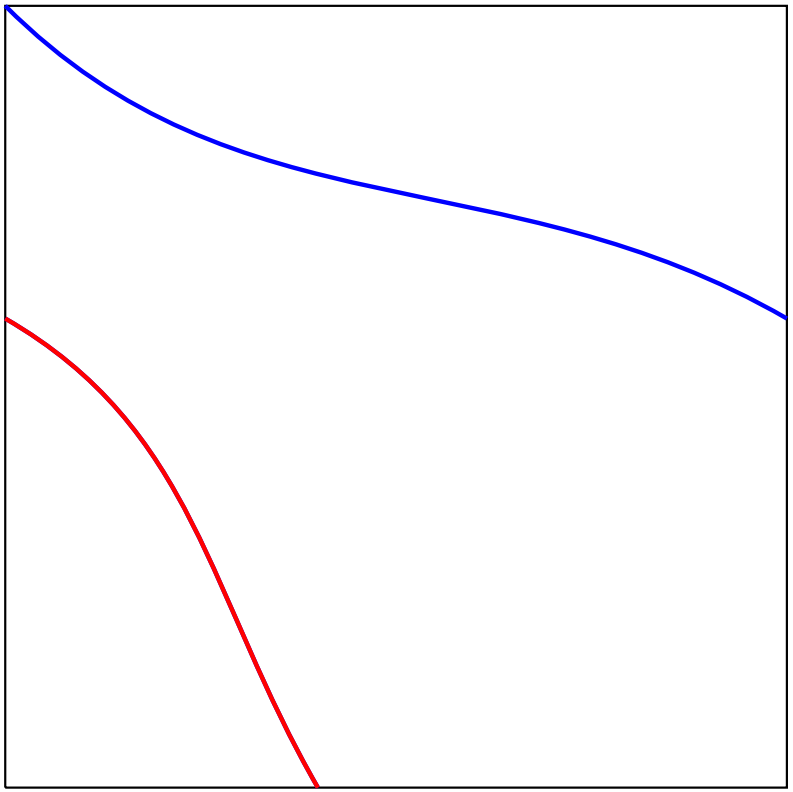

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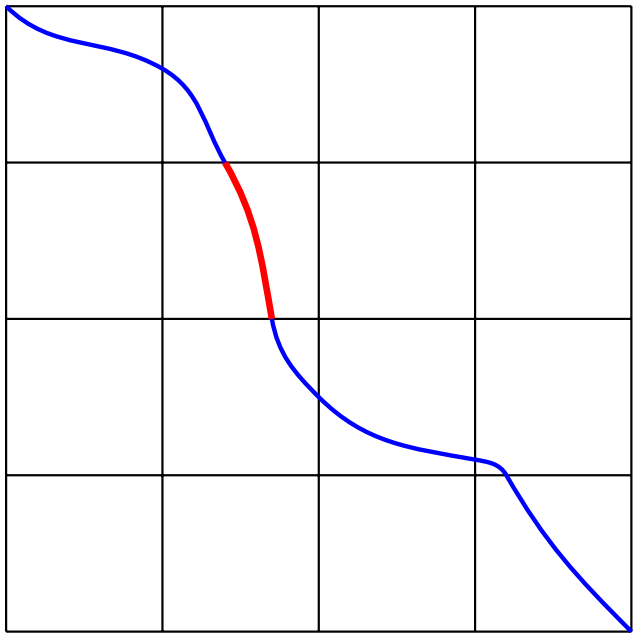

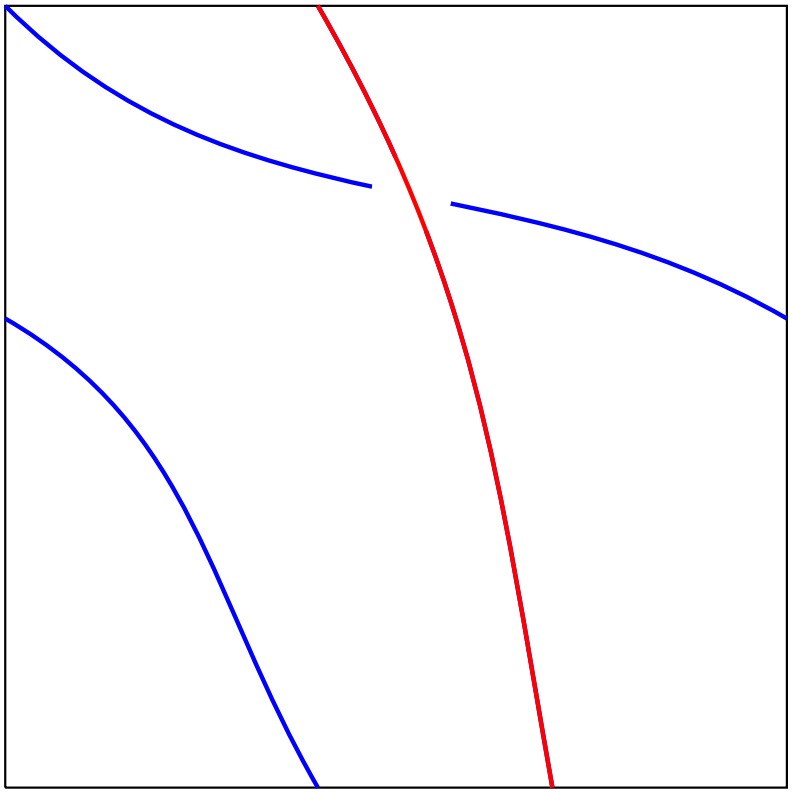

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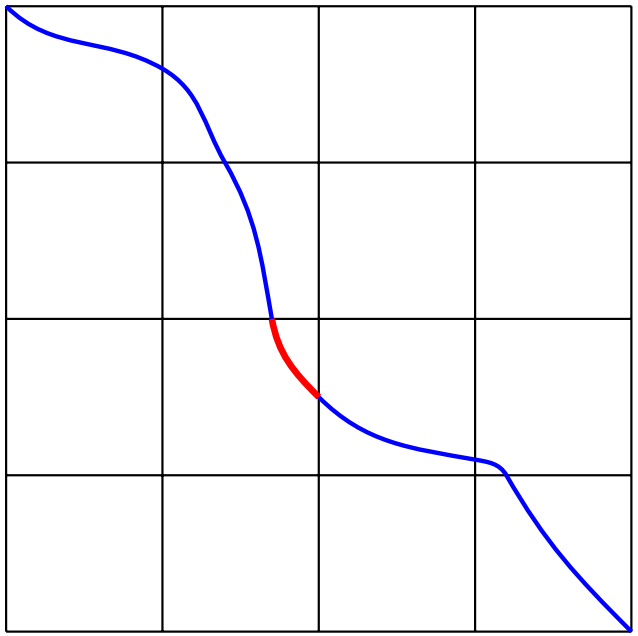

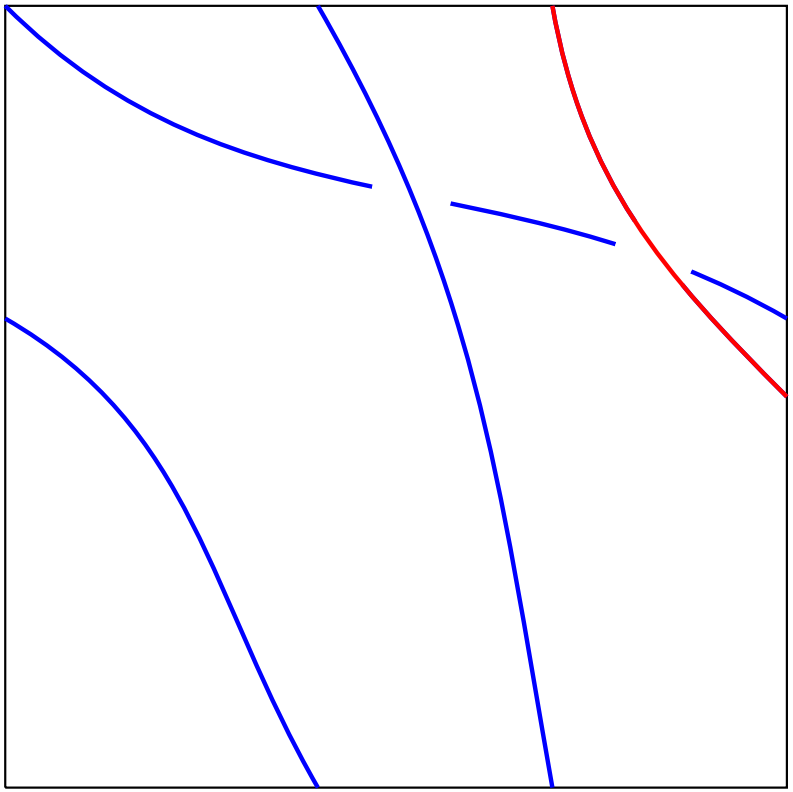

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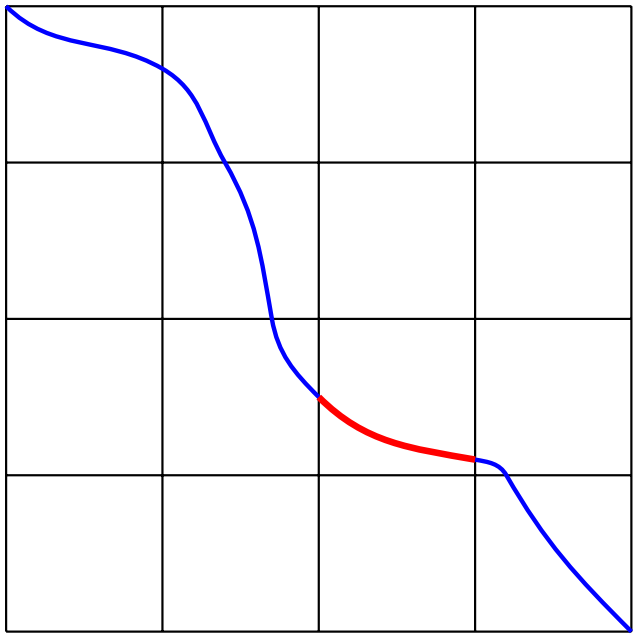

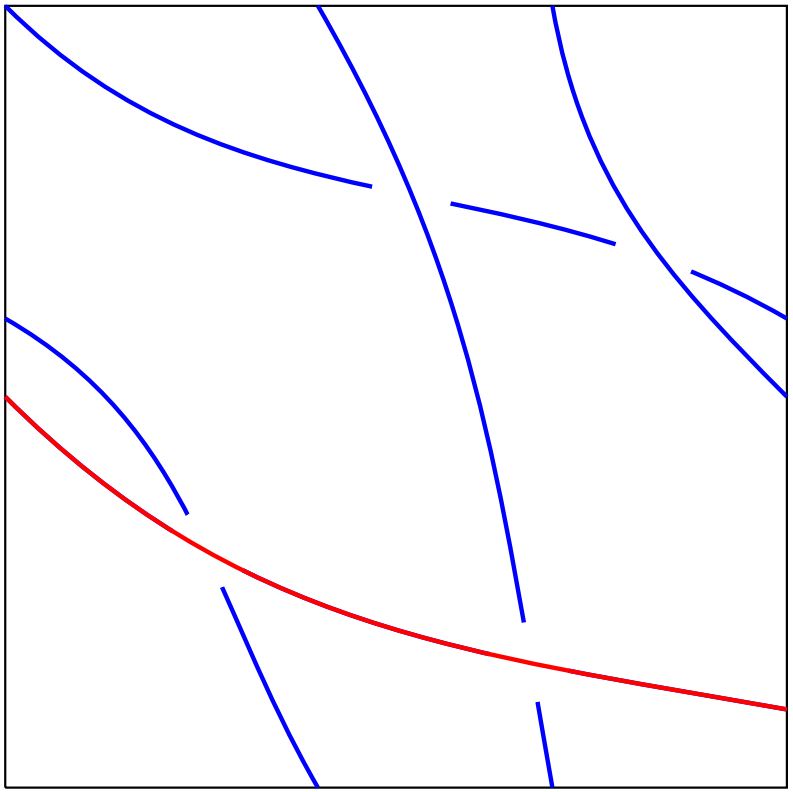

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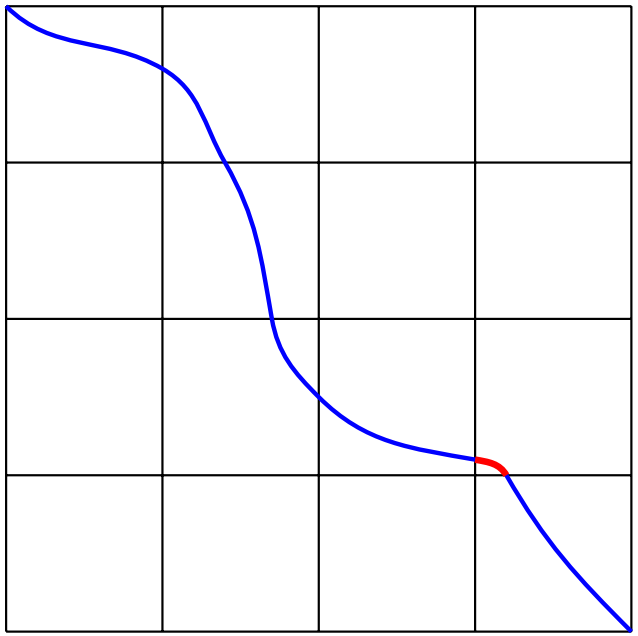
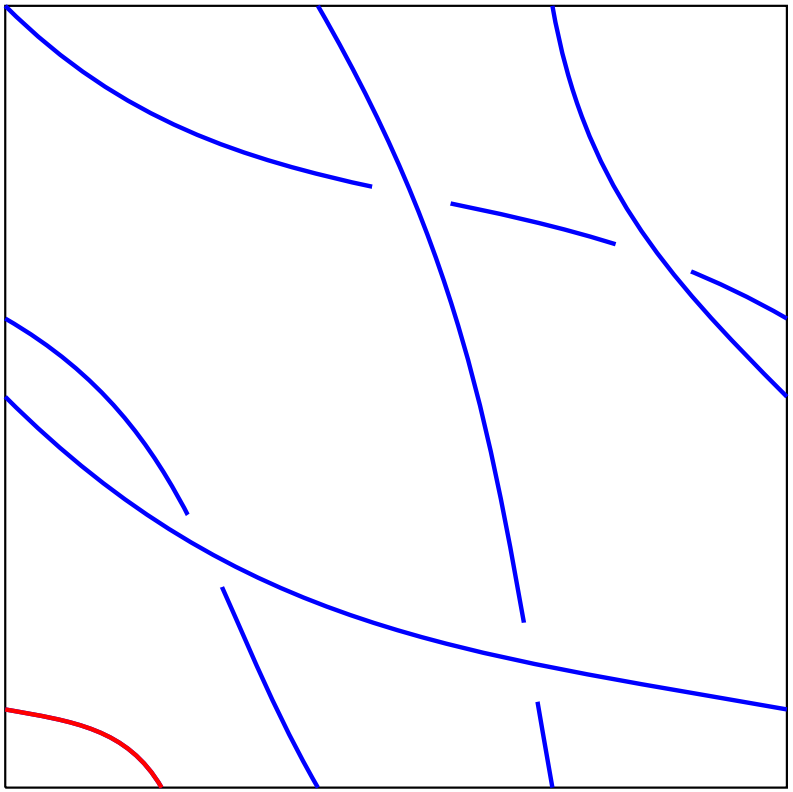
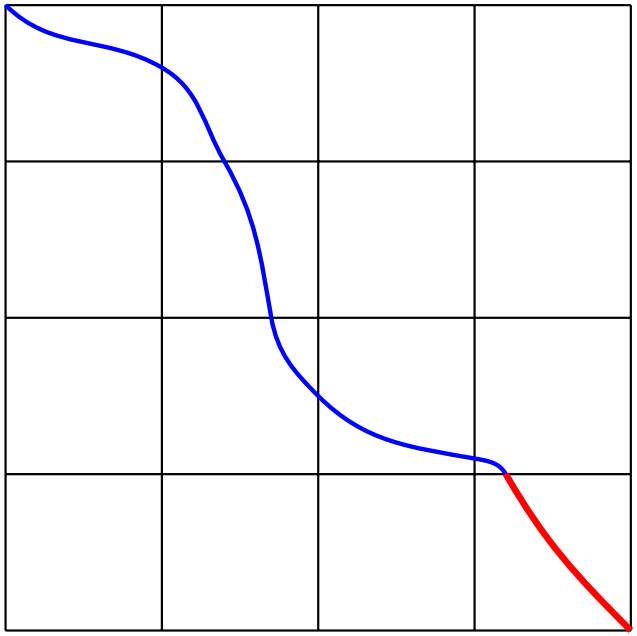
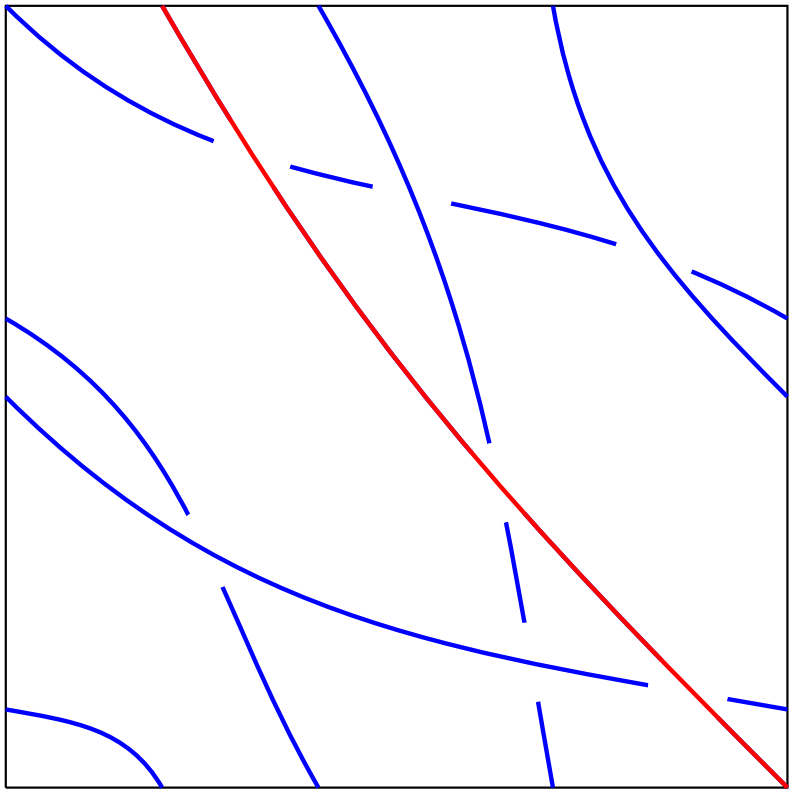




Monotone knots of Galashin-Lam





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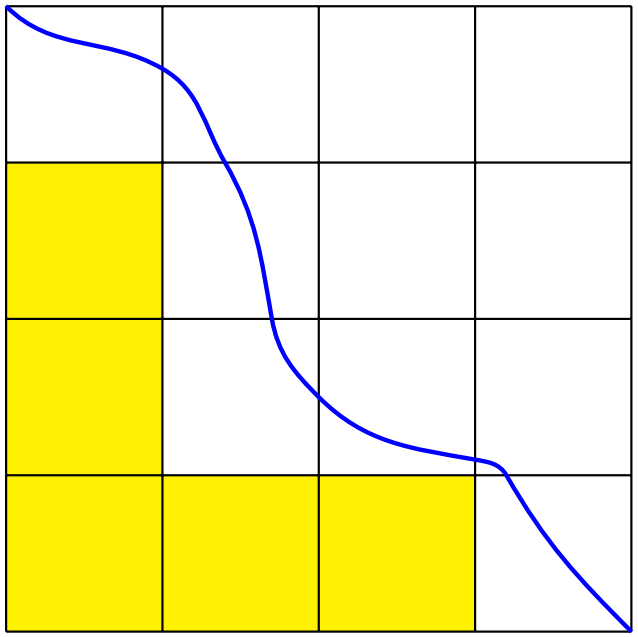
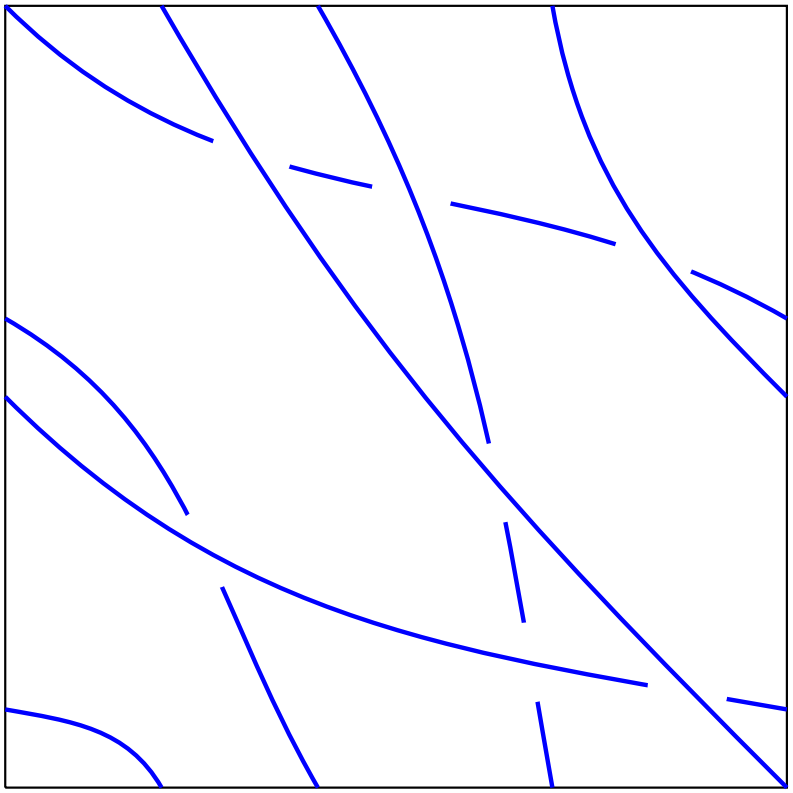
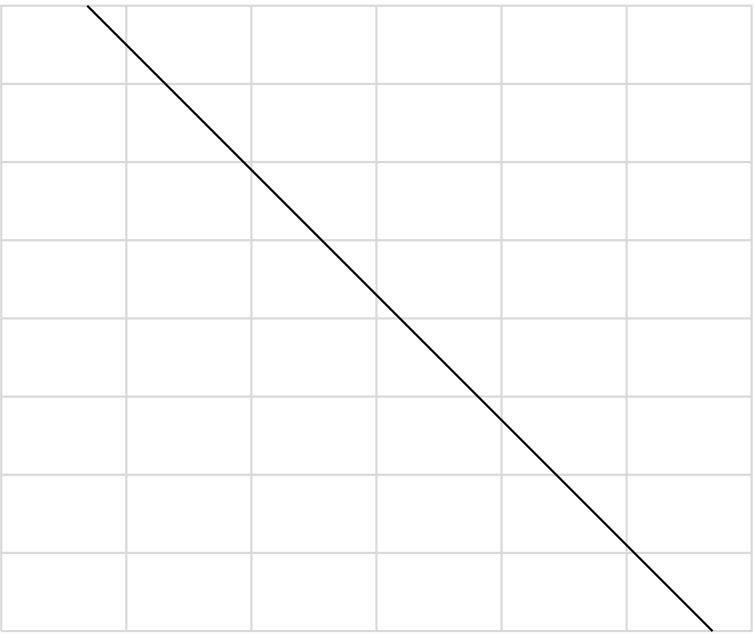


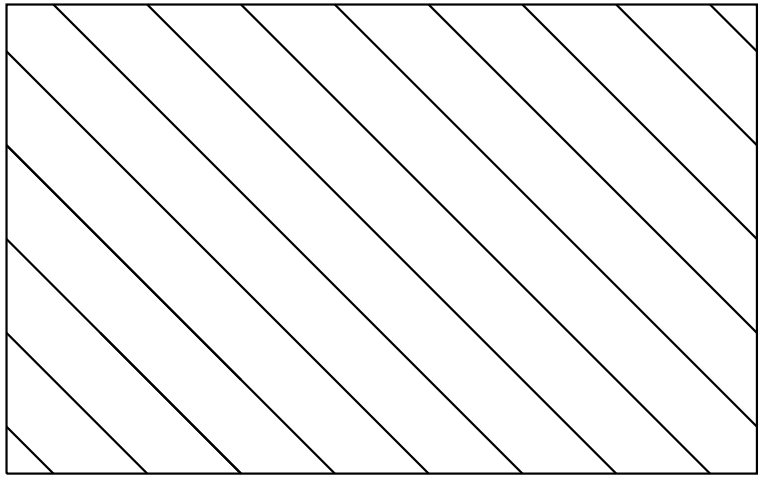


Theorem (Galashin-Lam)

Up to an isotopy, the knot only depends on the partition under the curve.

Monotone Knots of Triangular Partitions





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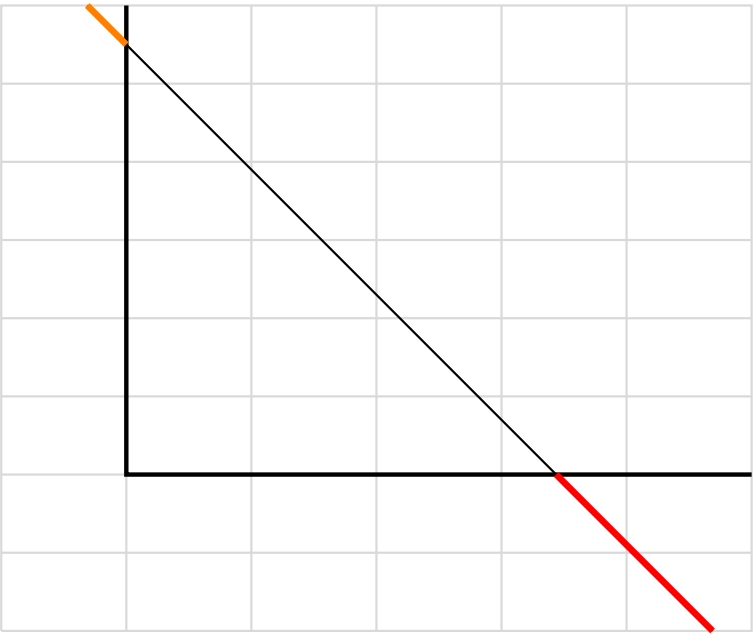

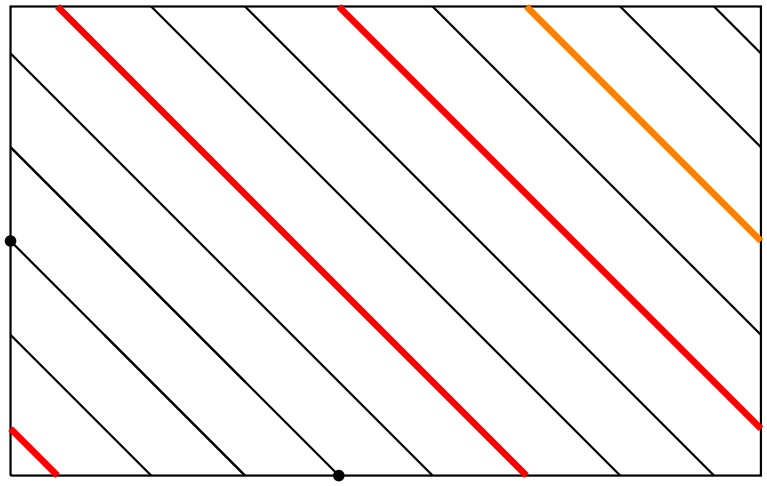

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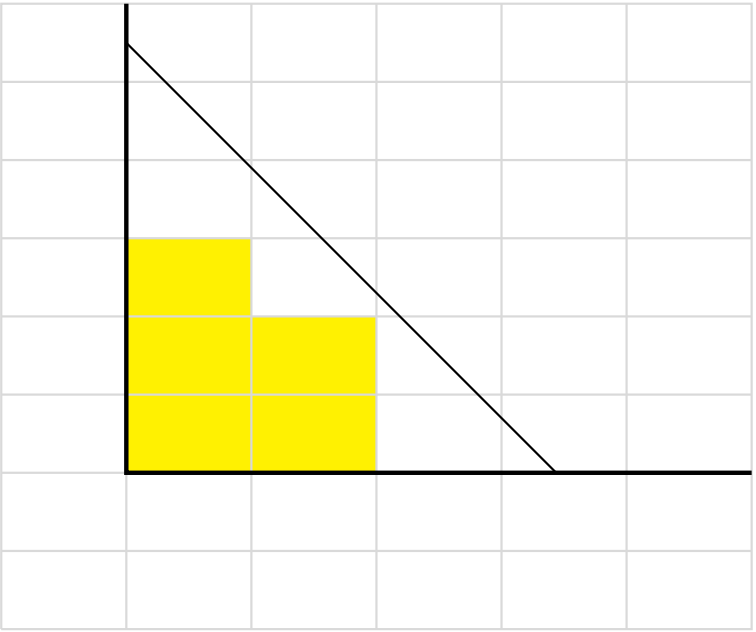
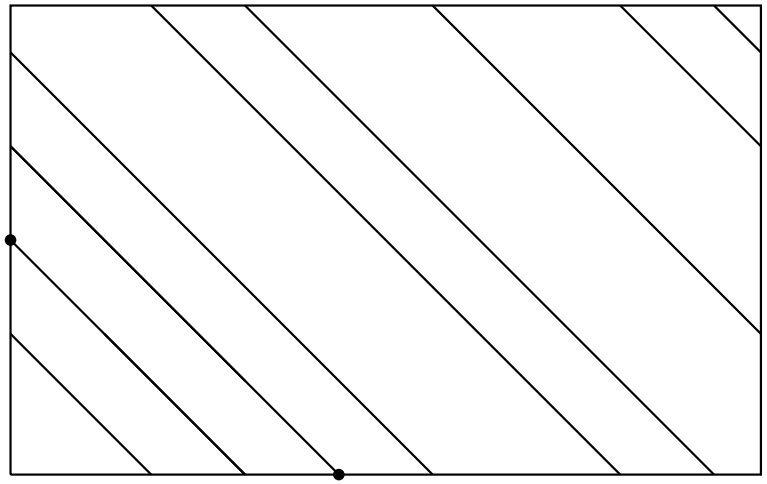
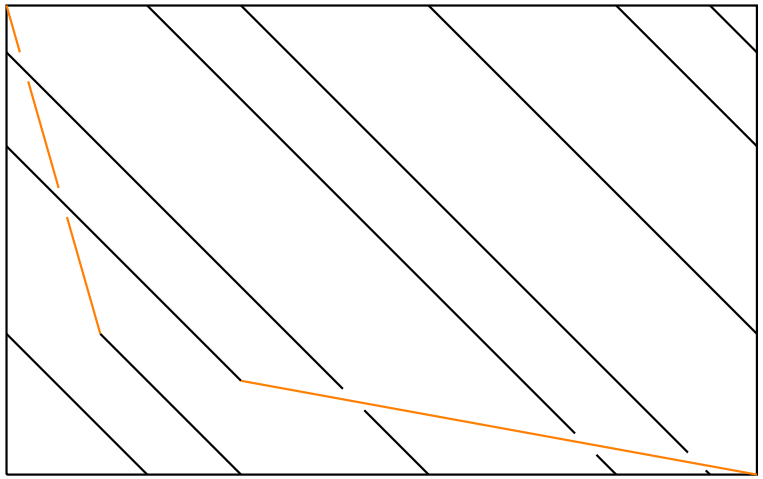
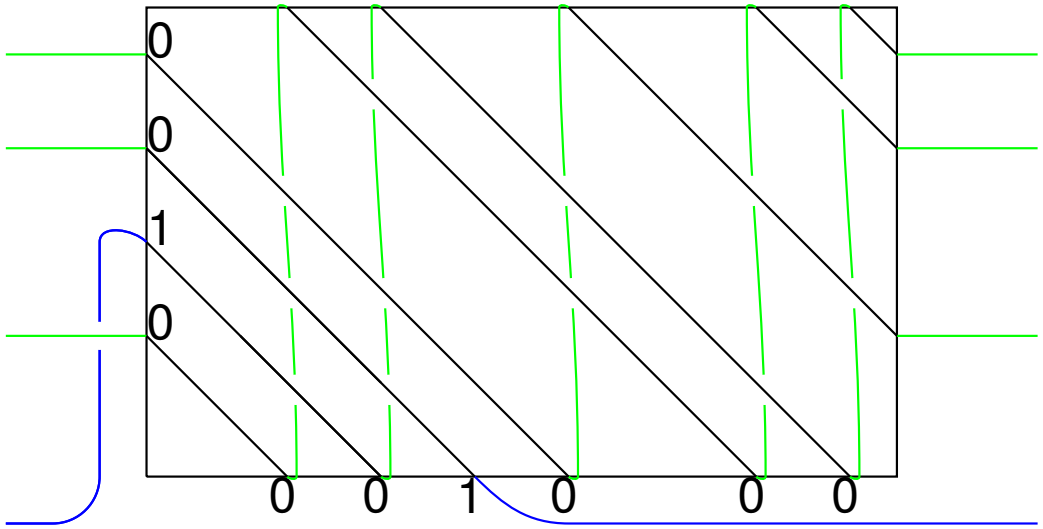




Monotone Knots of Triangular Partitions



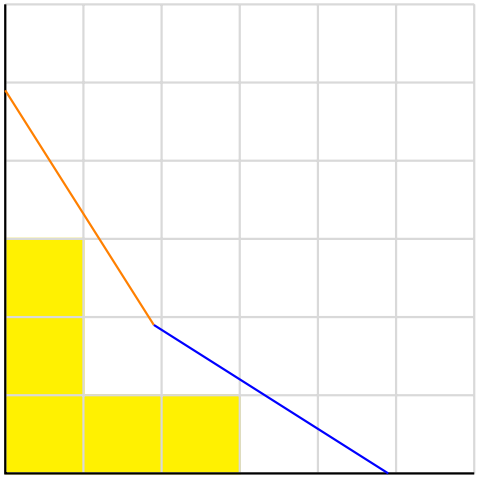
Monotone Knots of Triangular Partitions

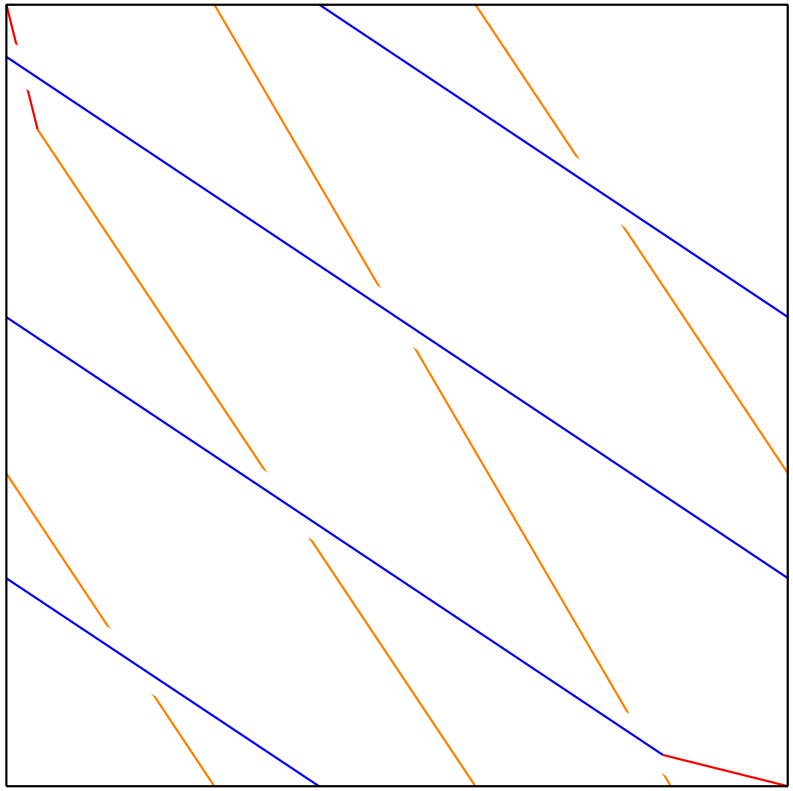


Theorem

Monotone knots of the triangular partitions are the shortcut torus knots.

Example: $(3, 1, 1)$





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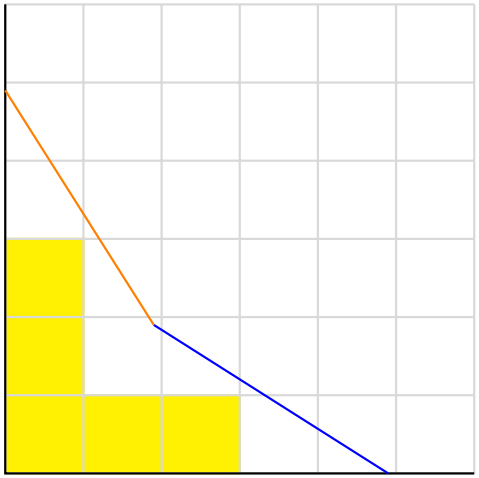

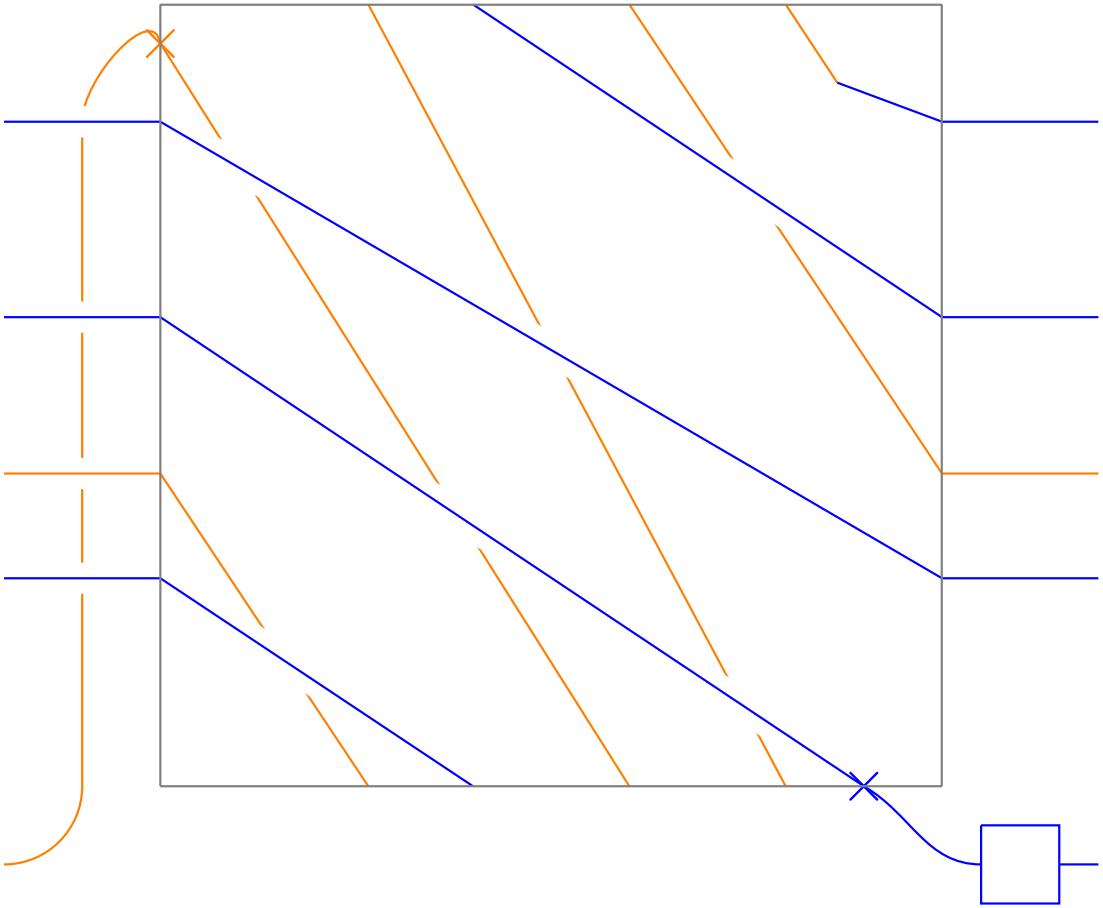

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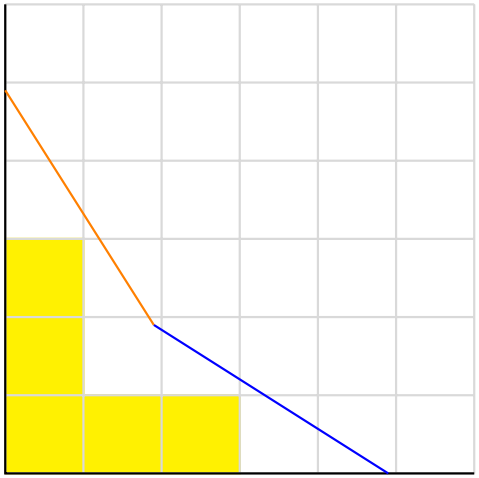

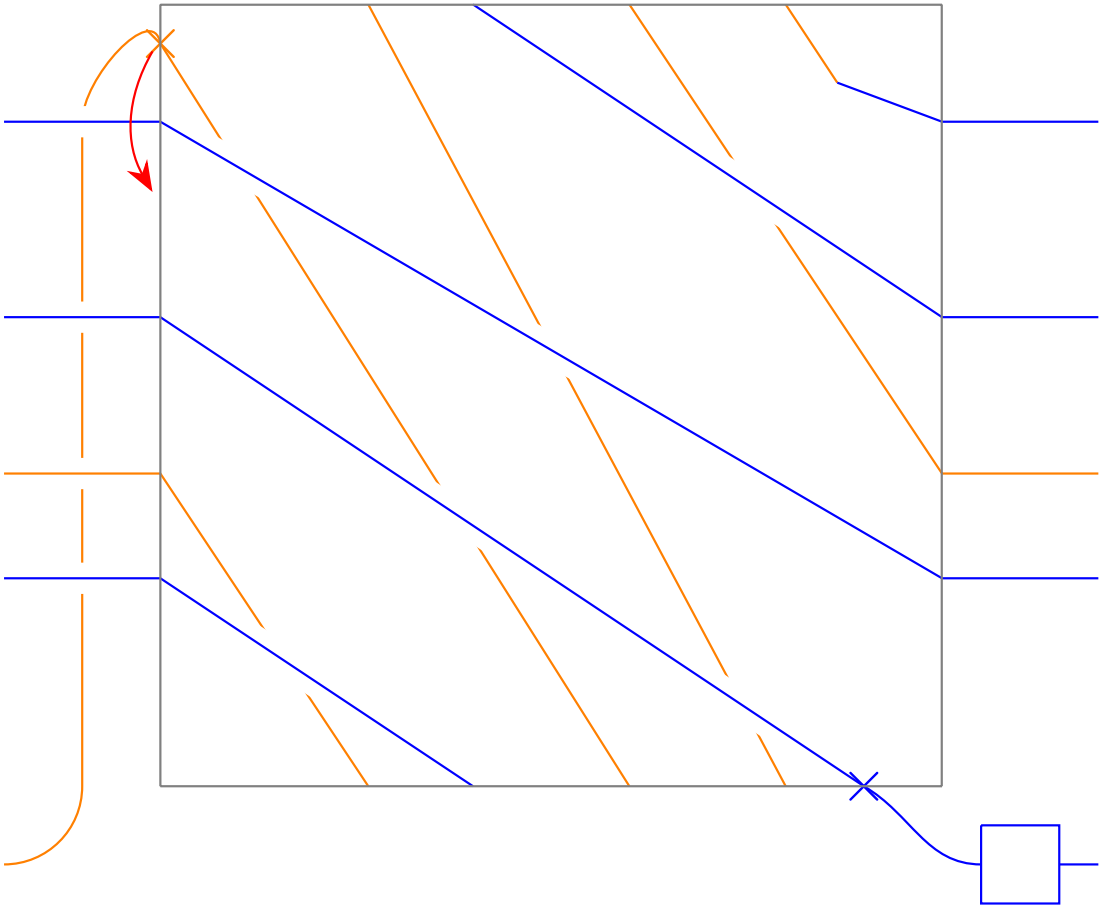

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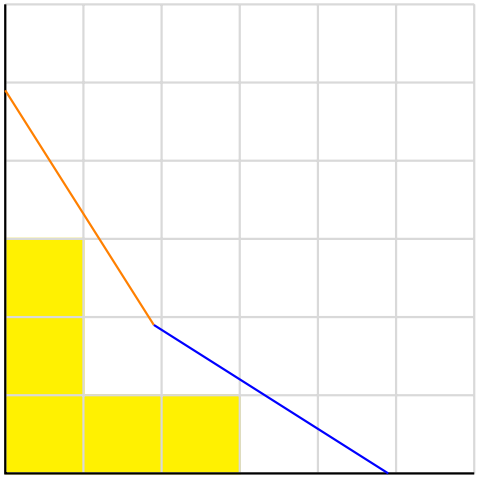

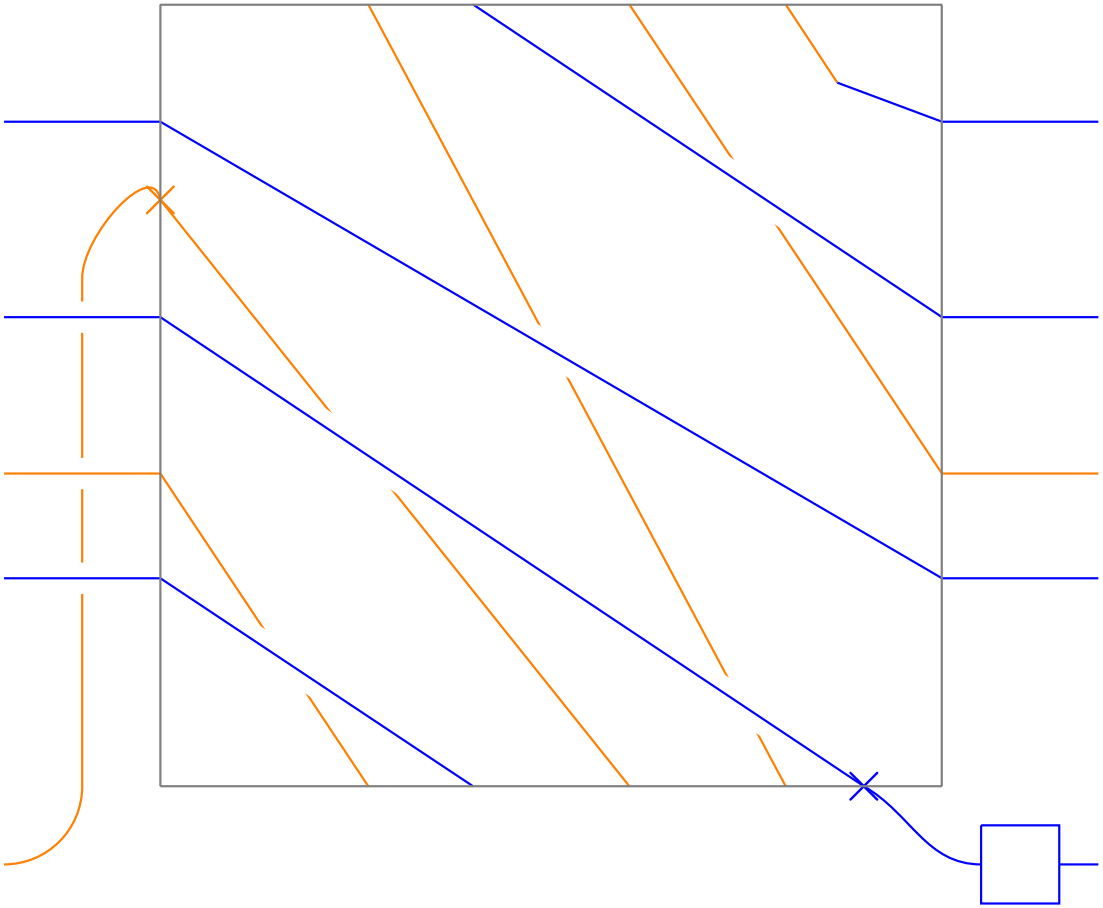

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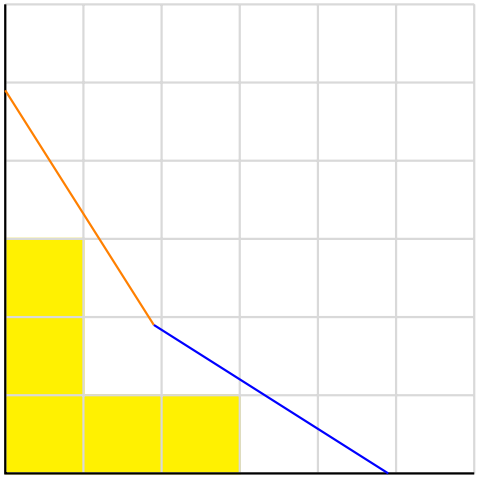

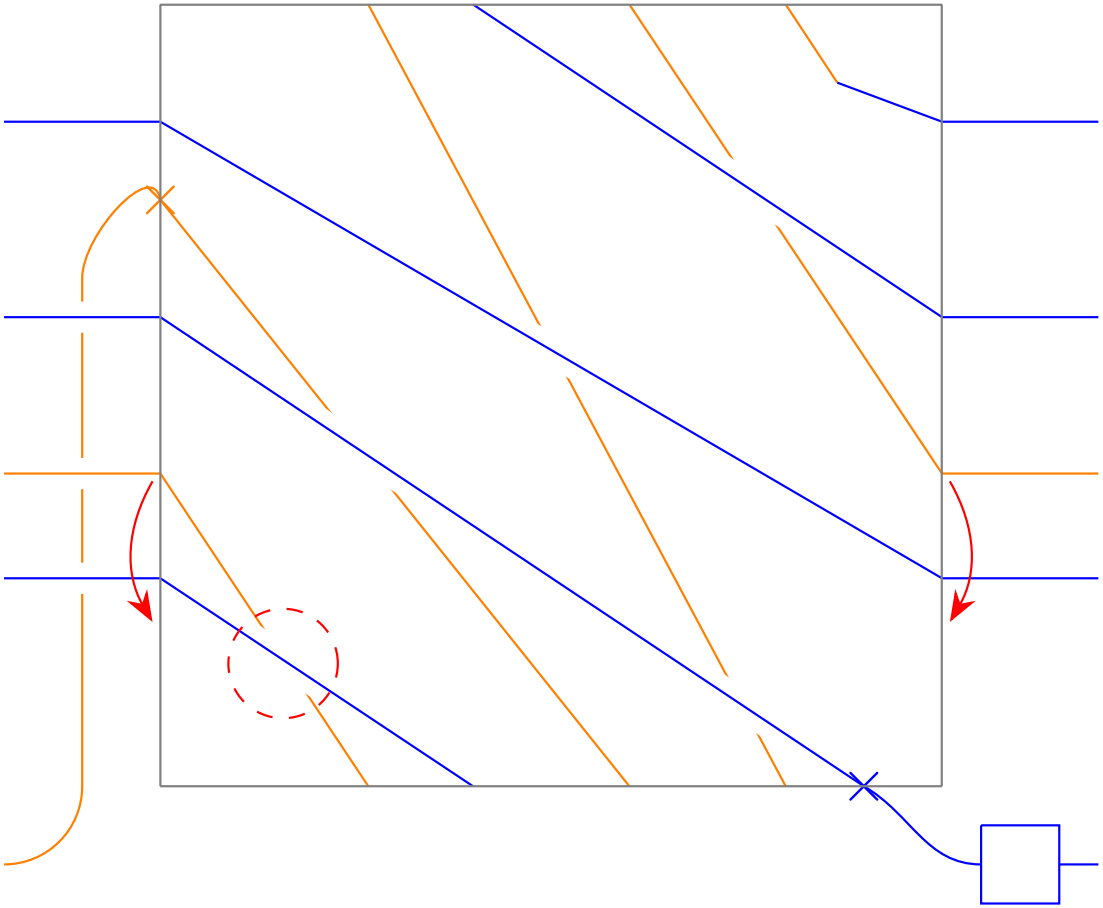

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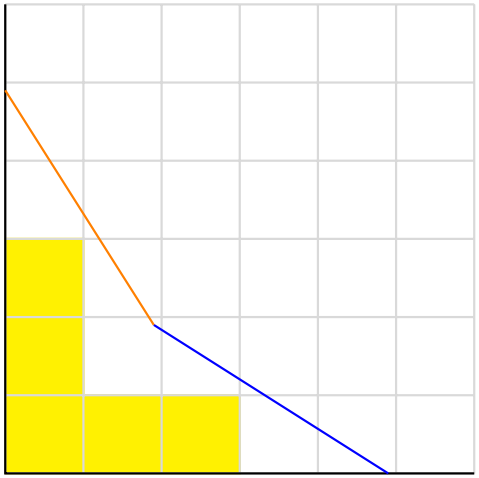

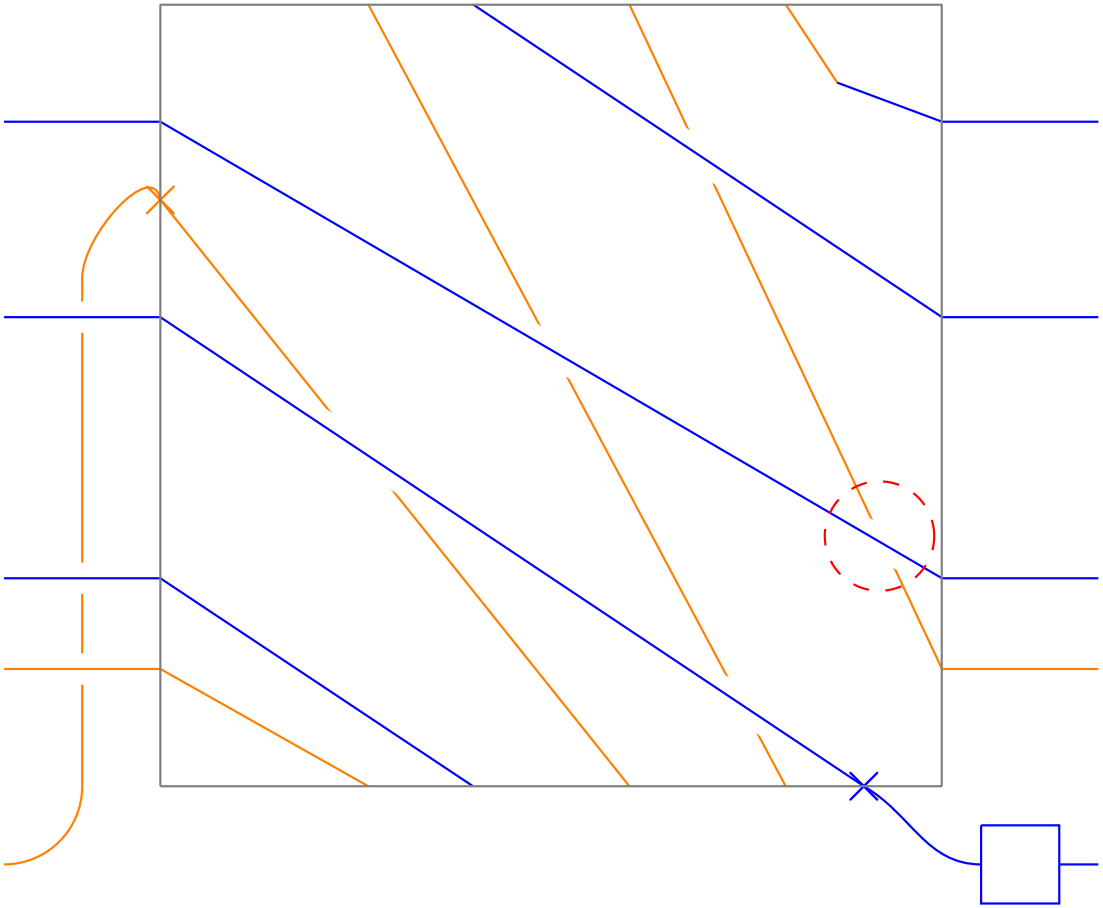

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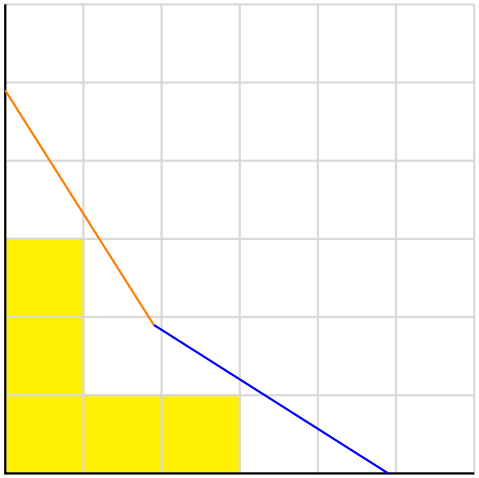

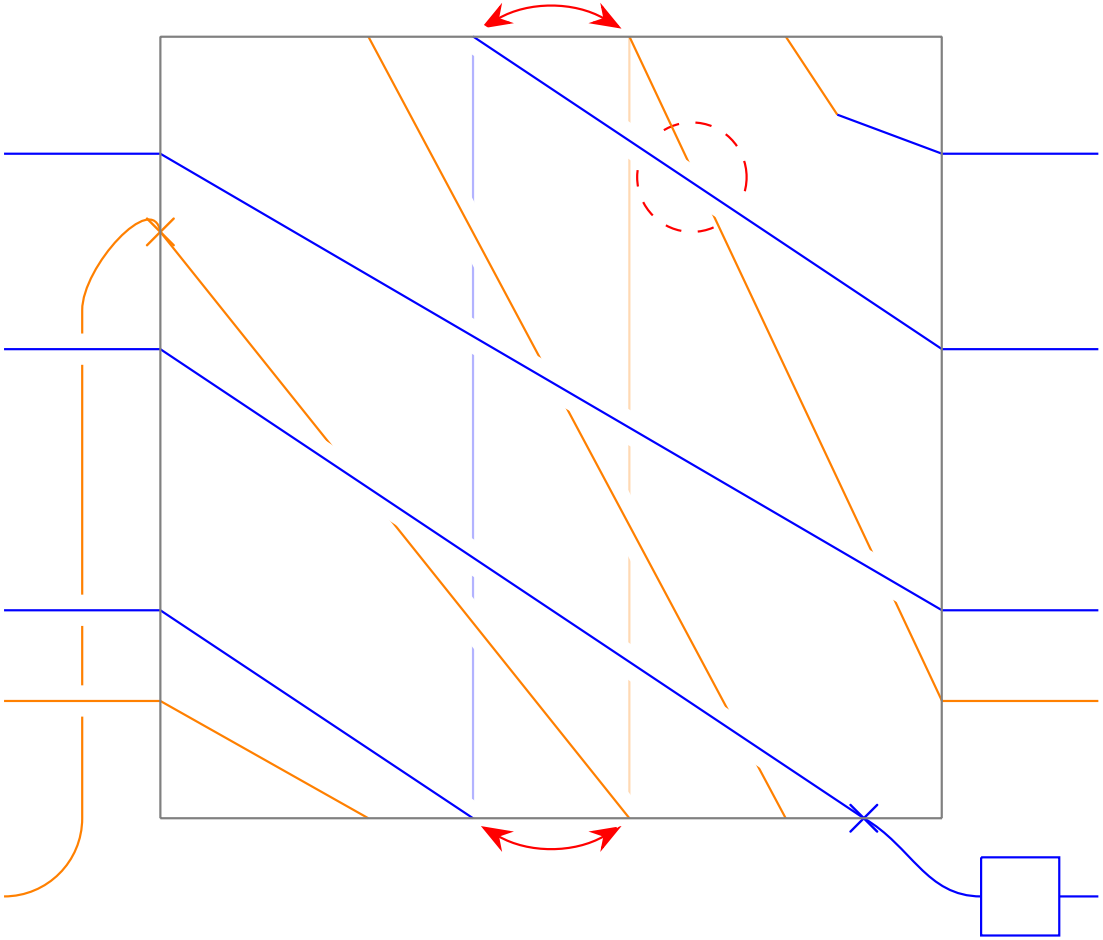

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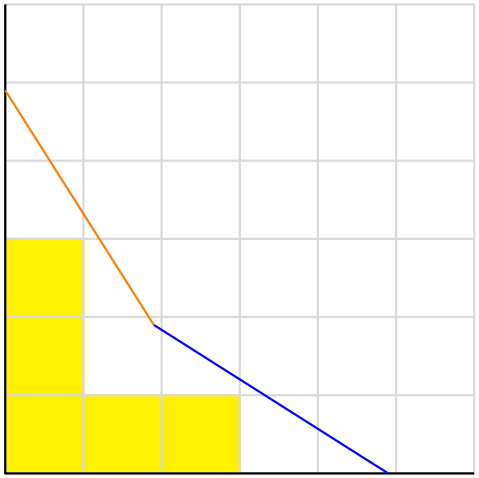

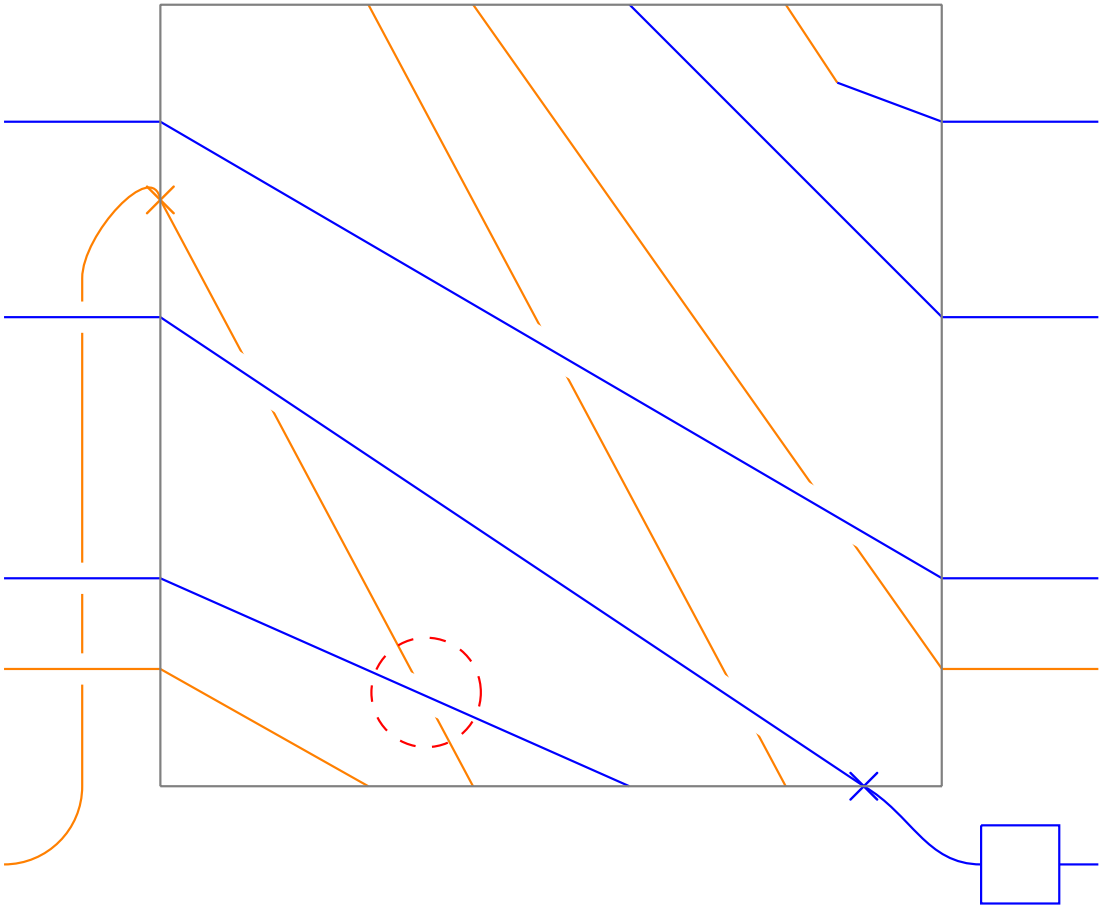

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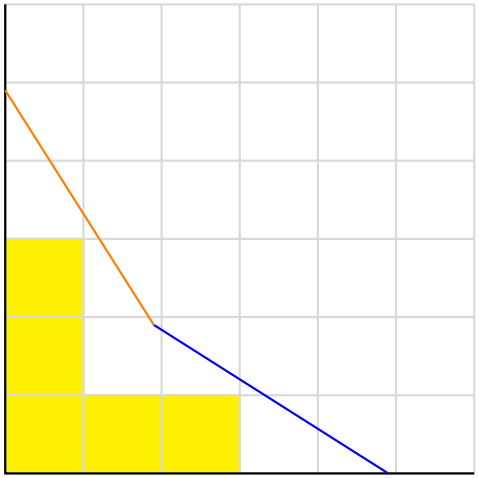

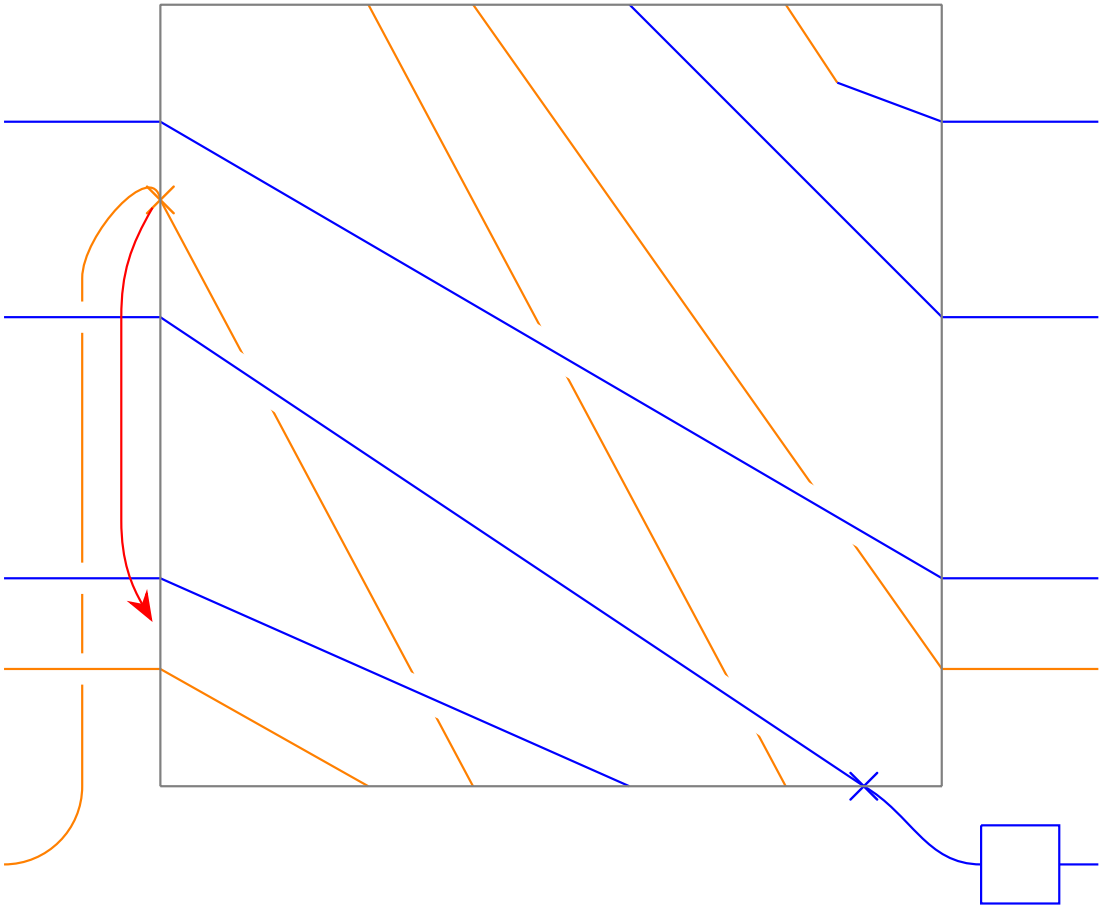

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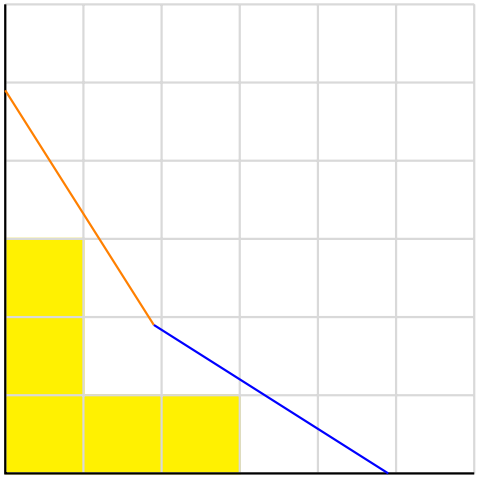

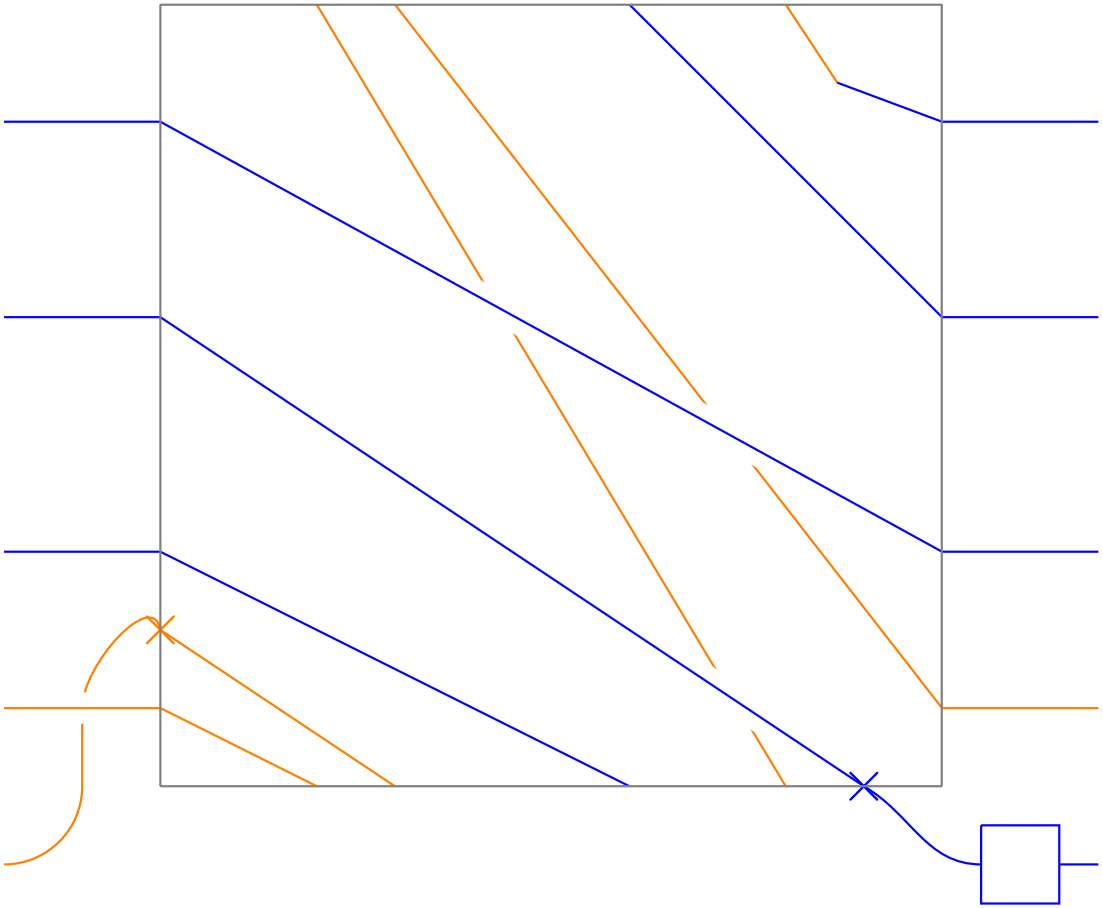

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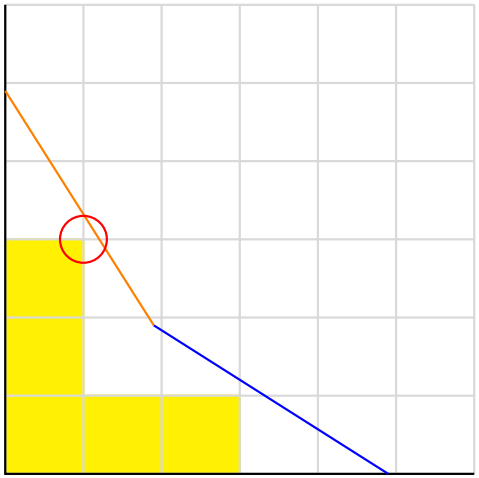

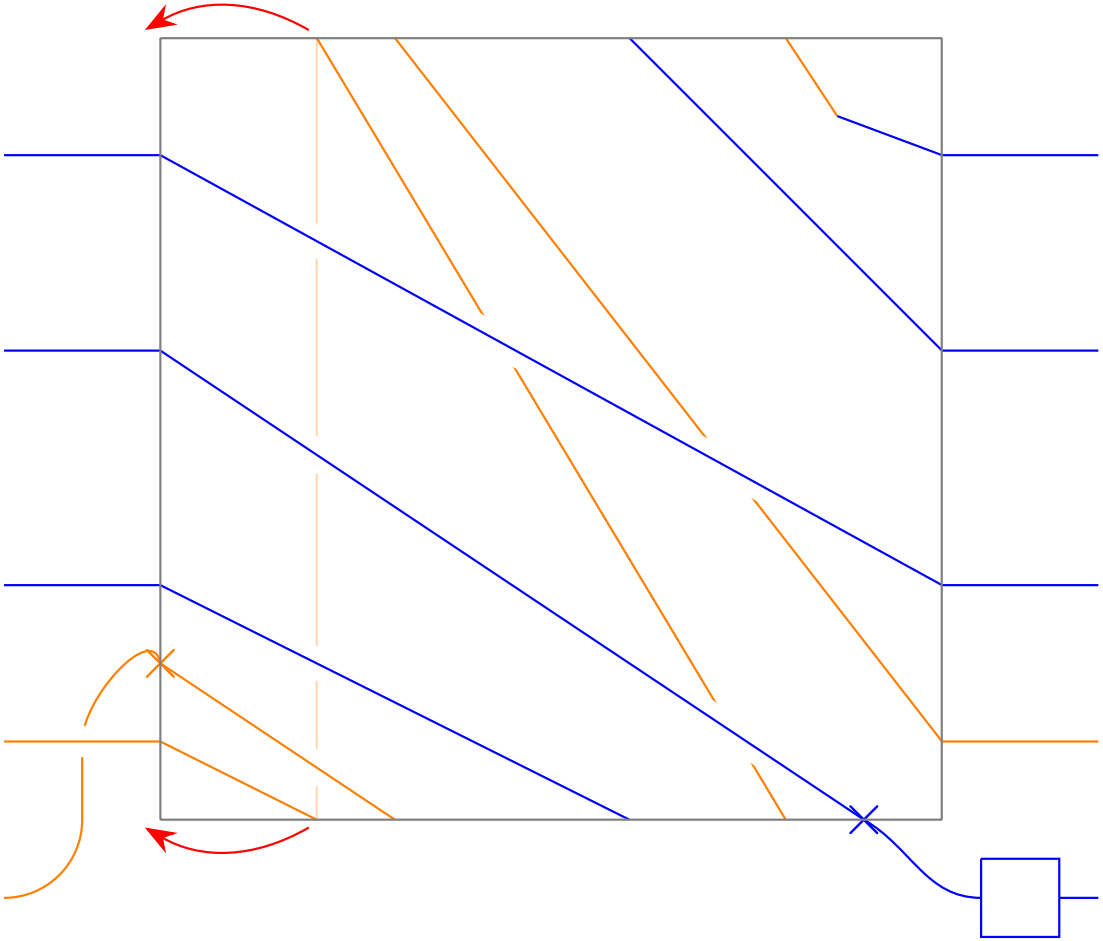

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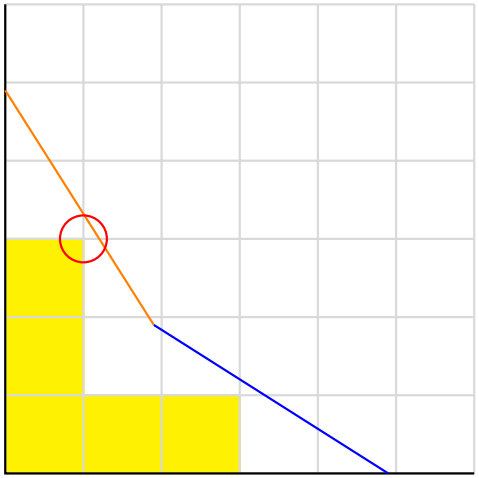

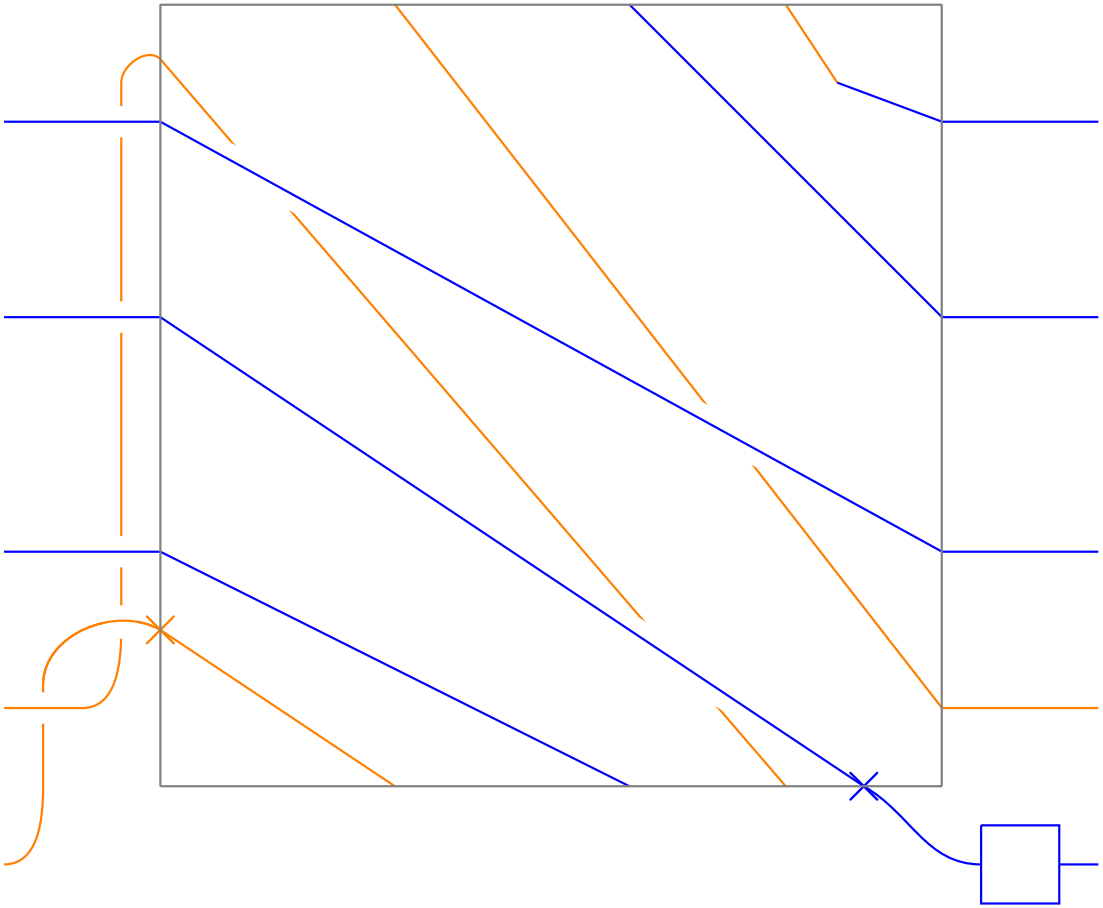

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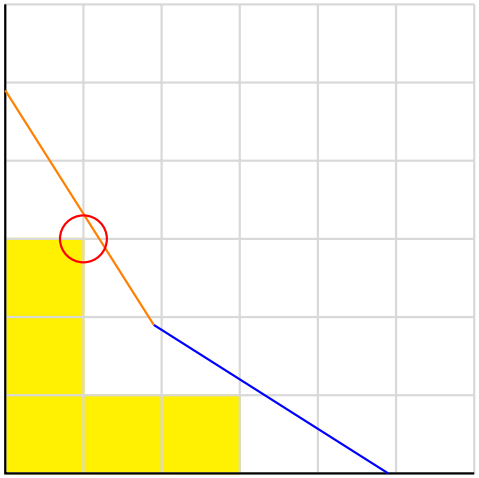

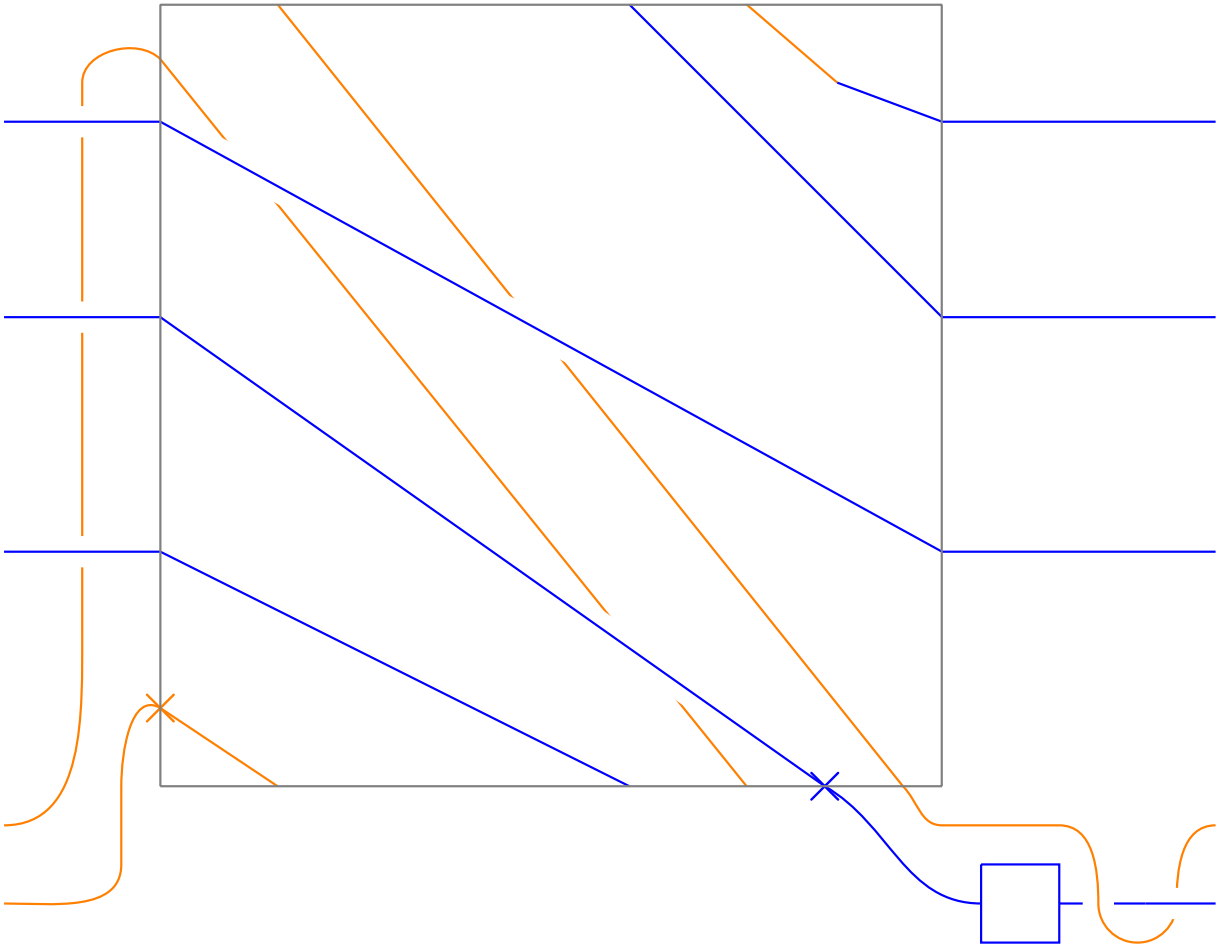

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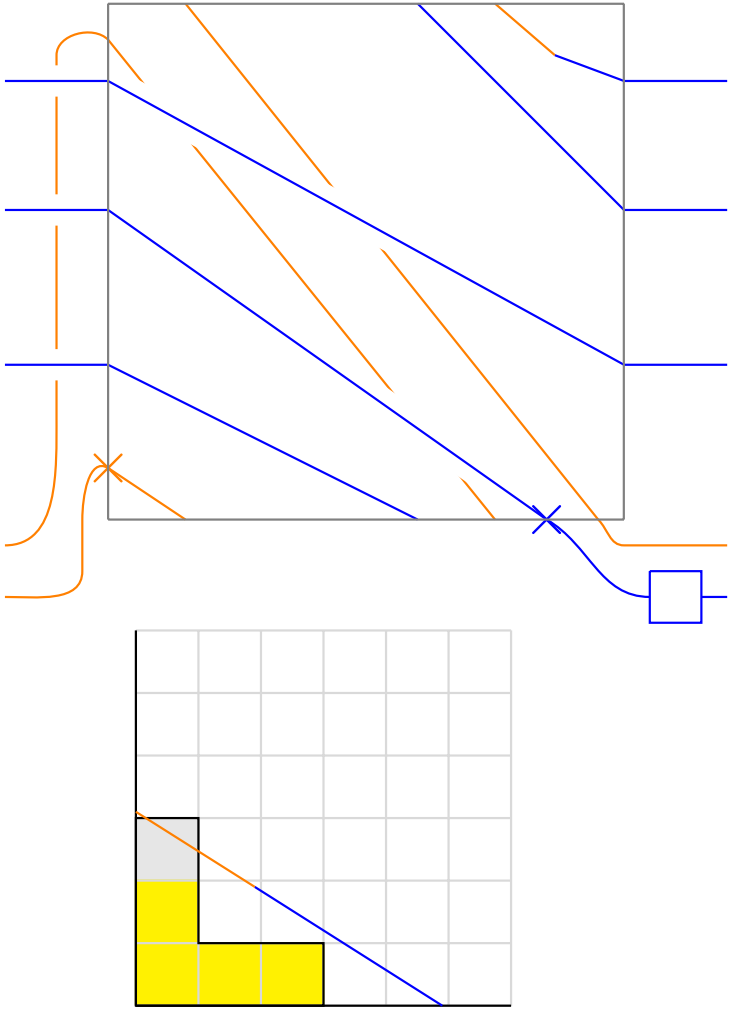

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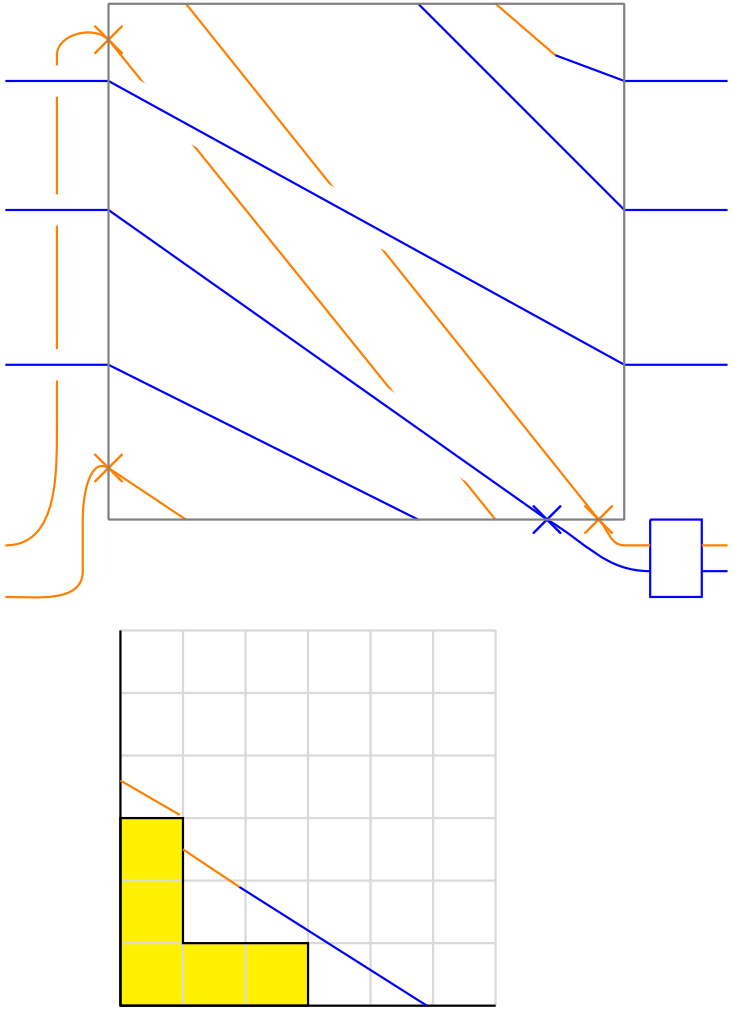

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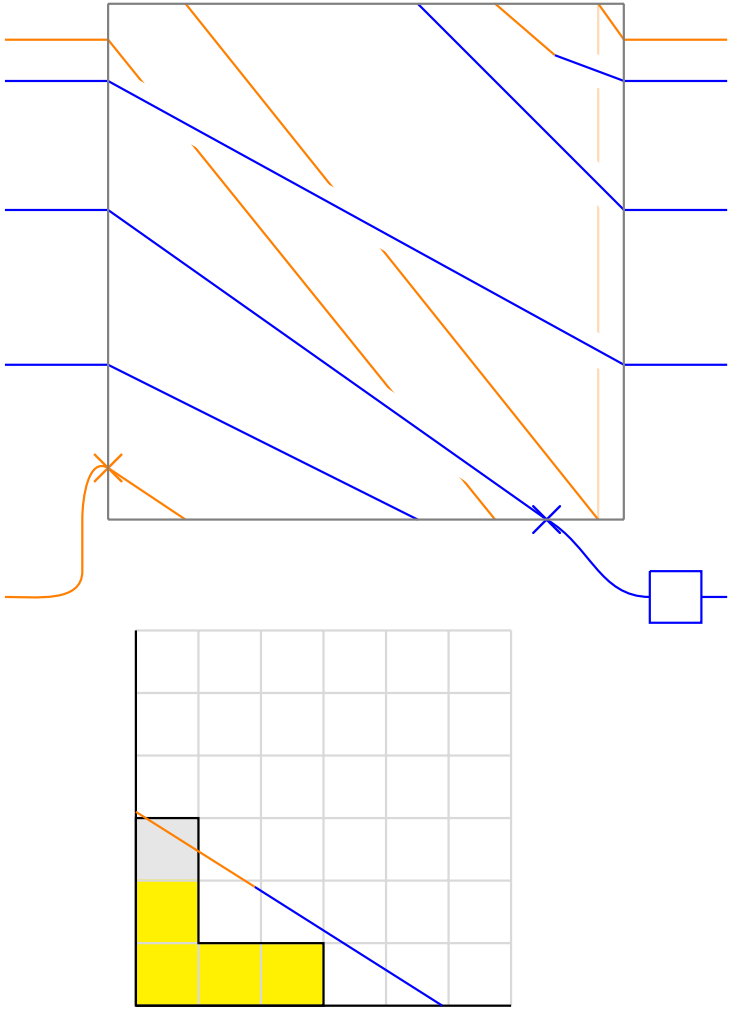

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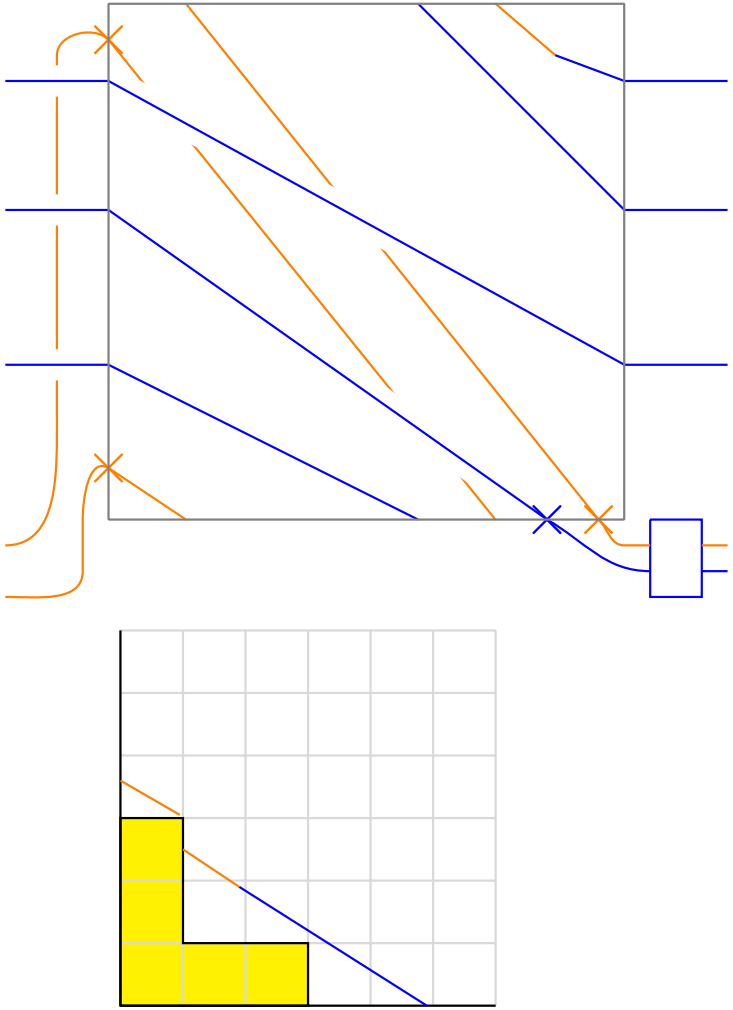

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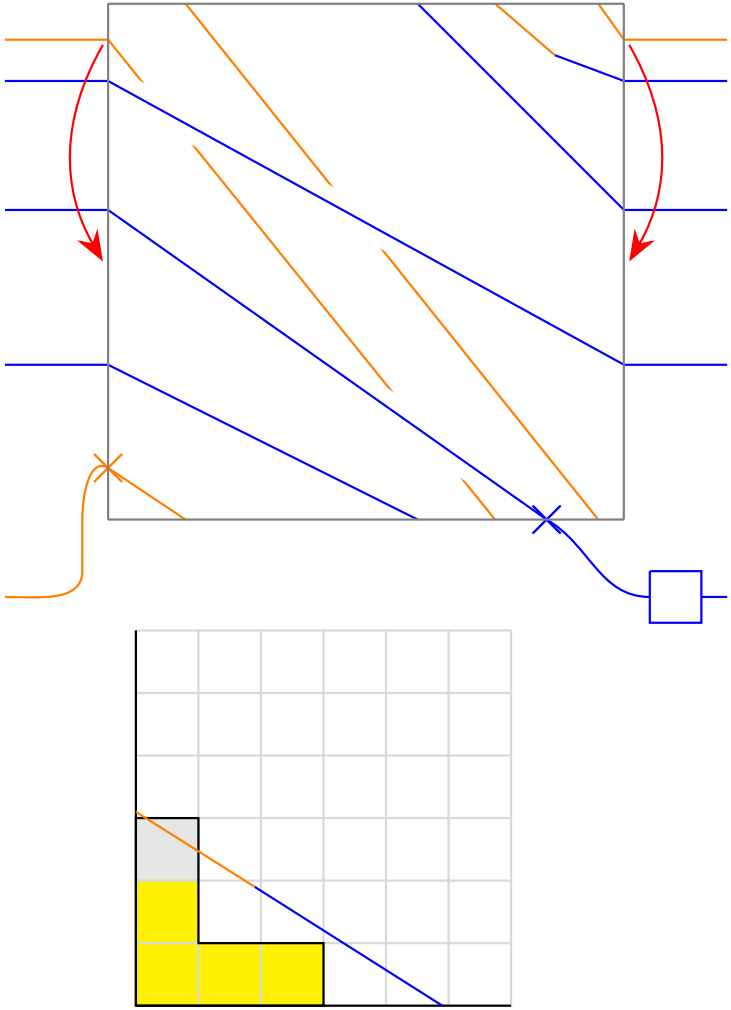

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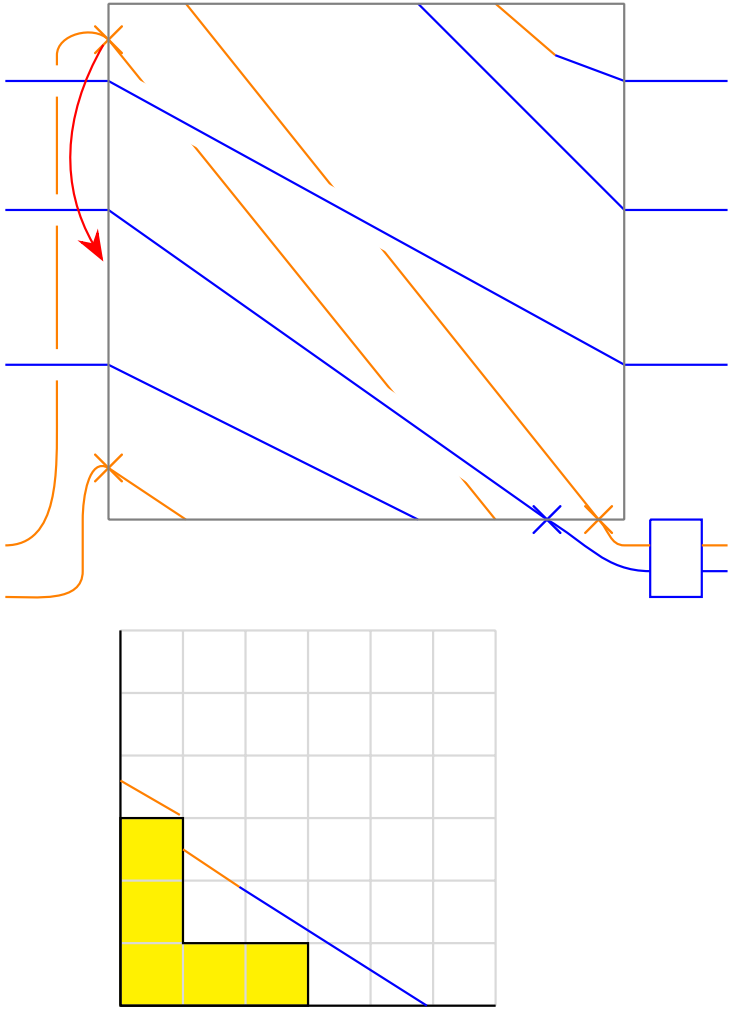

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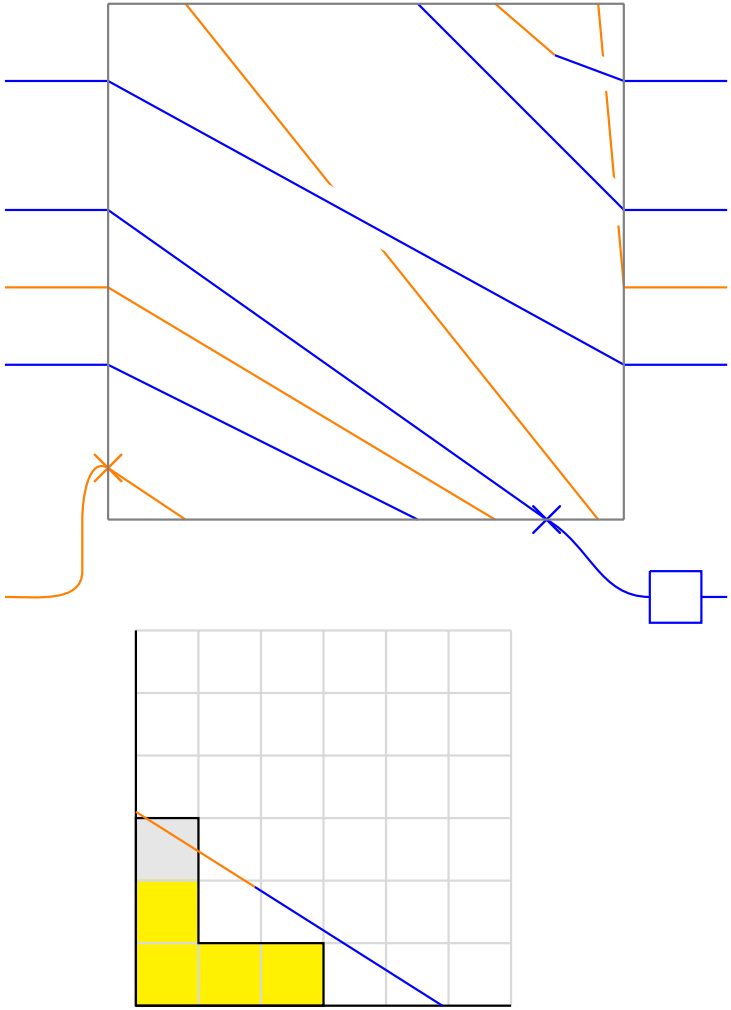

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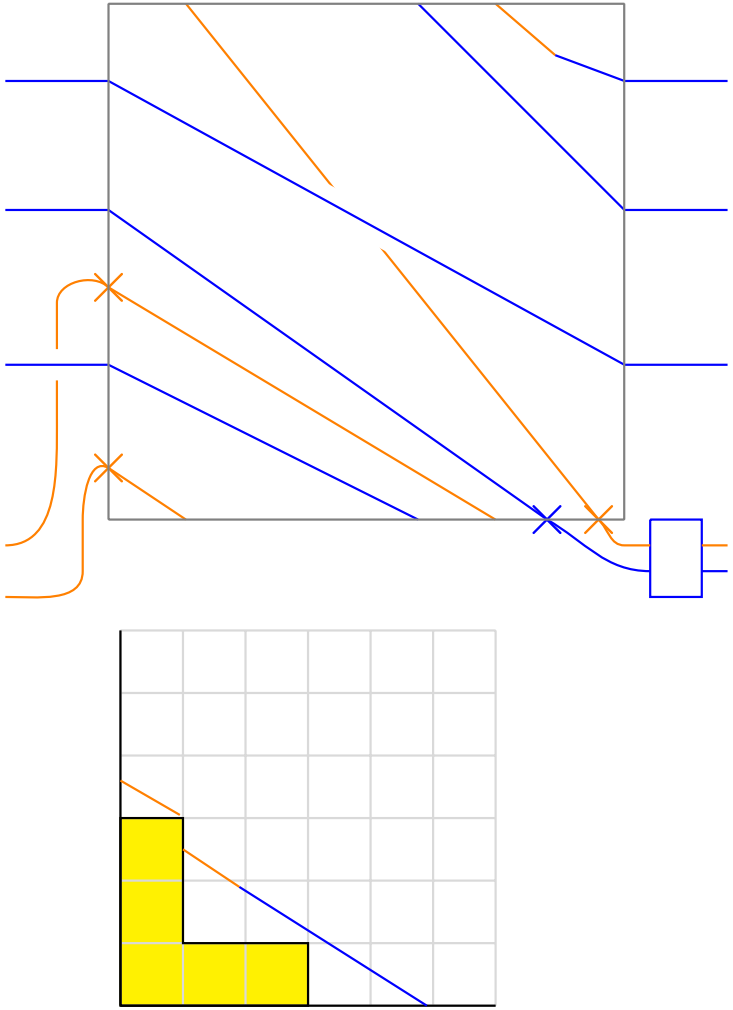

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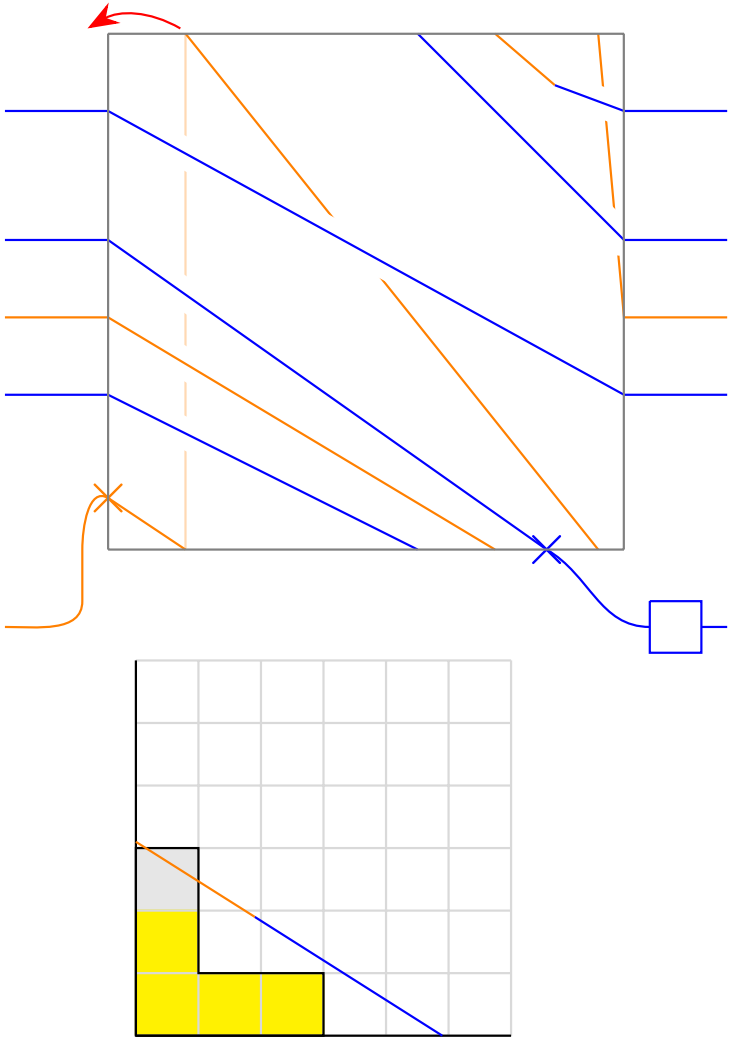

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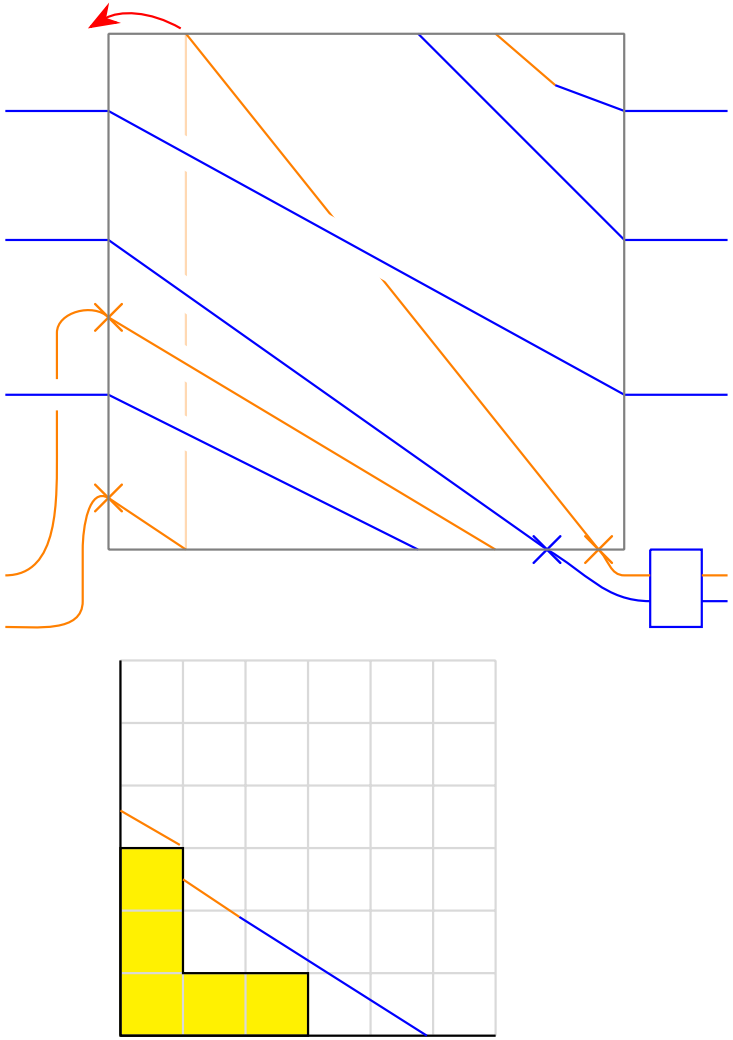

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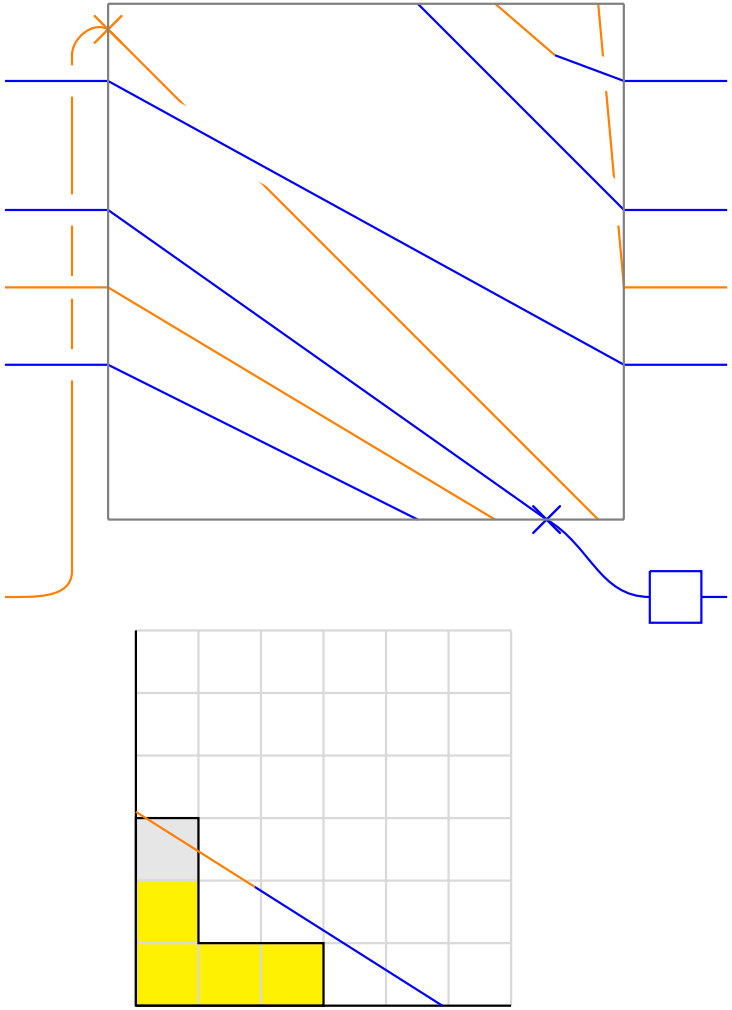

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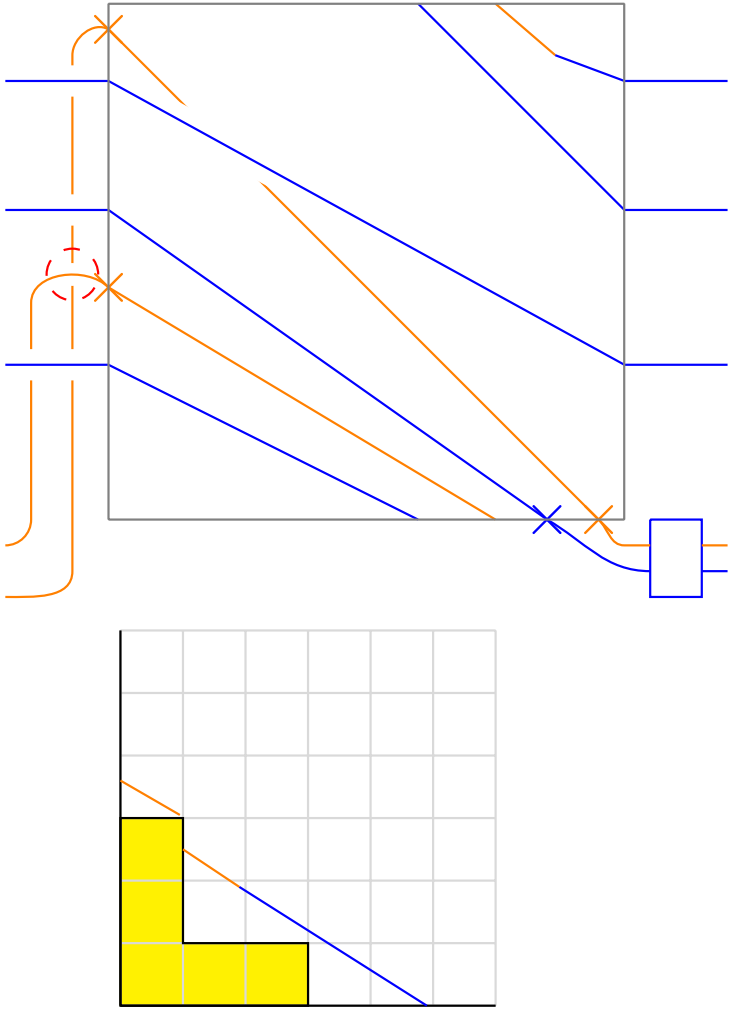

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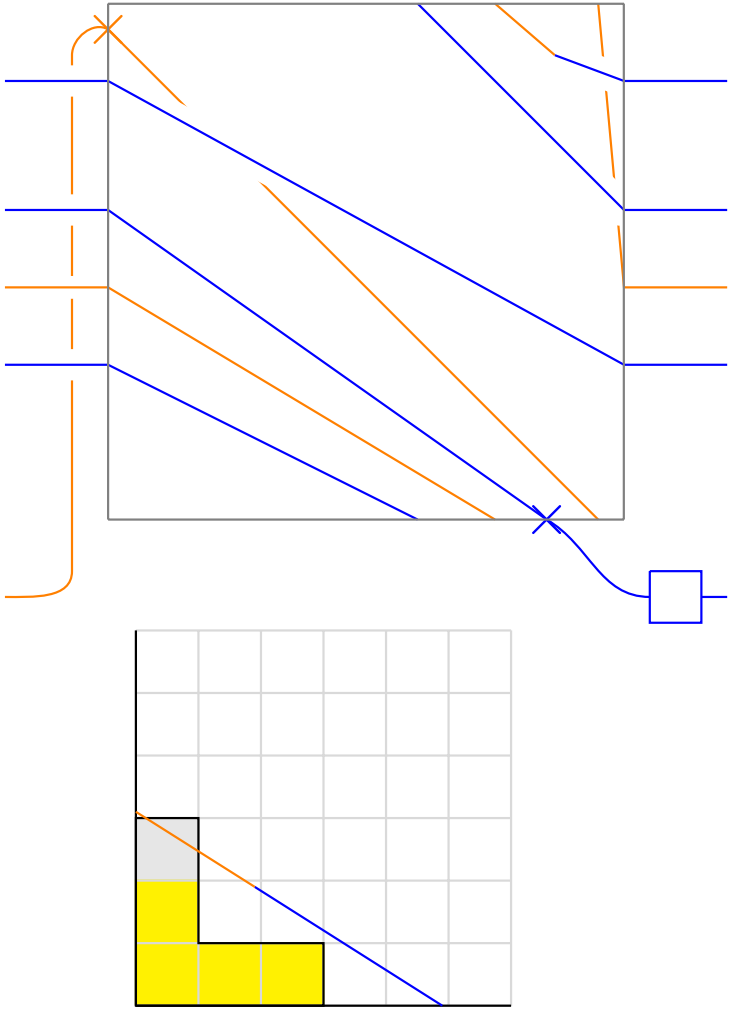

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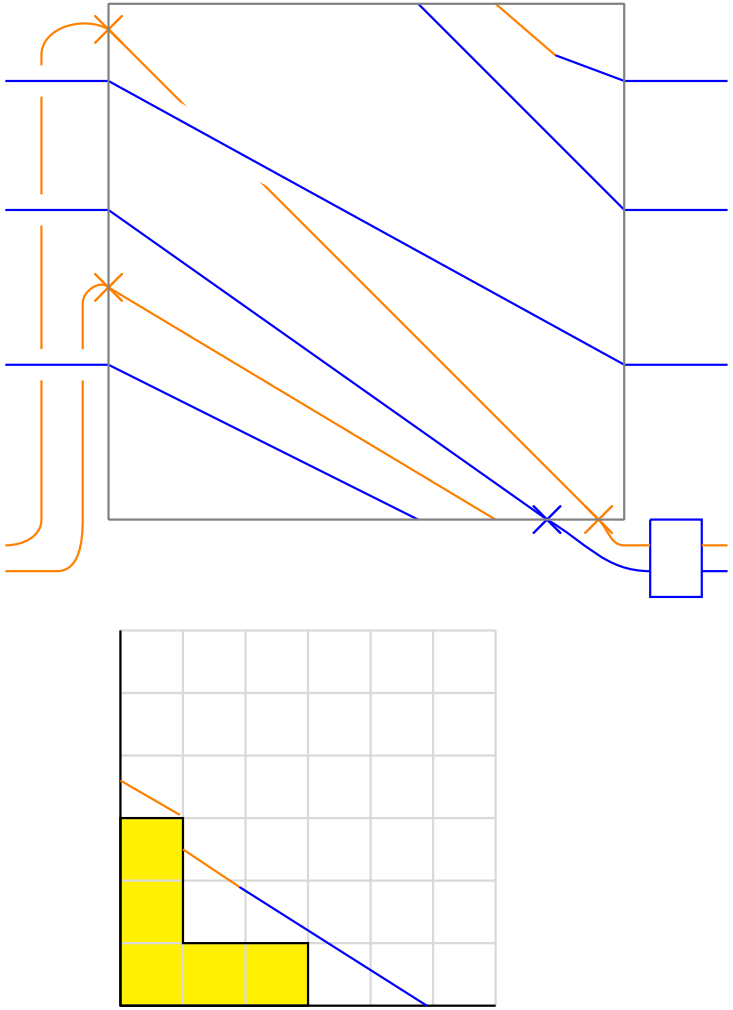

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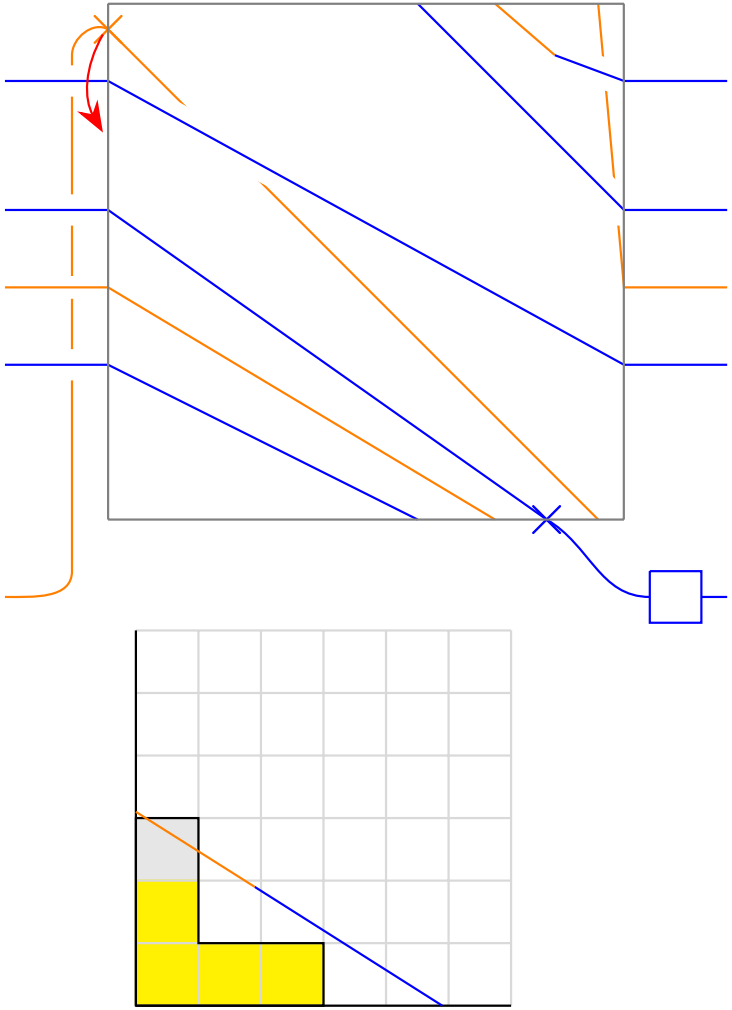

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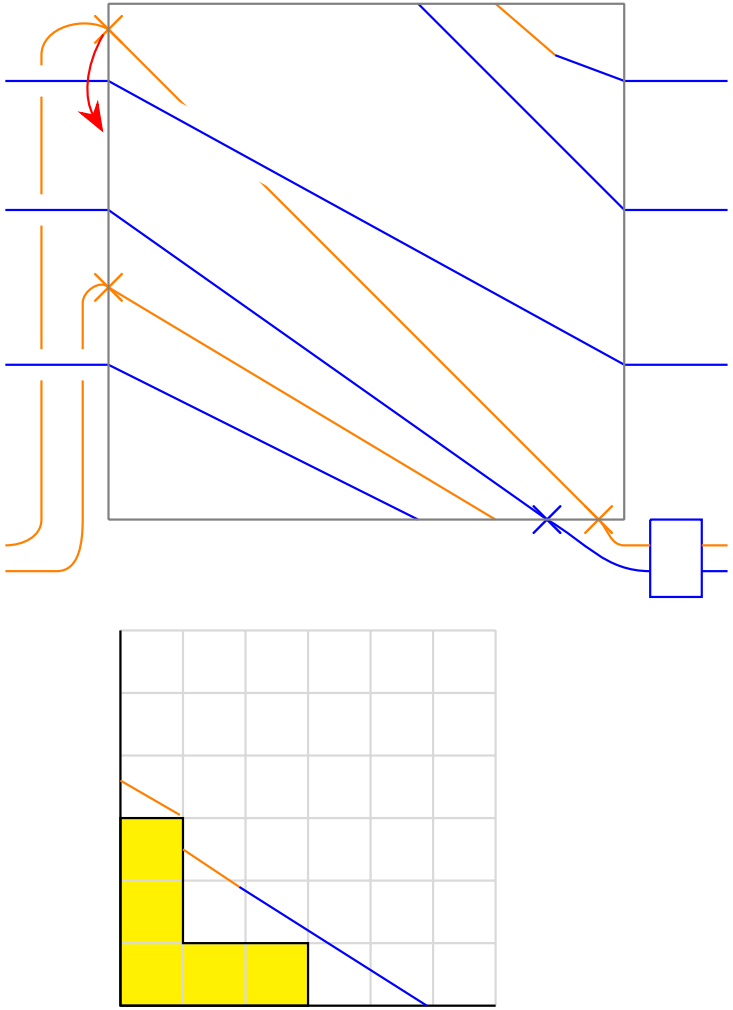

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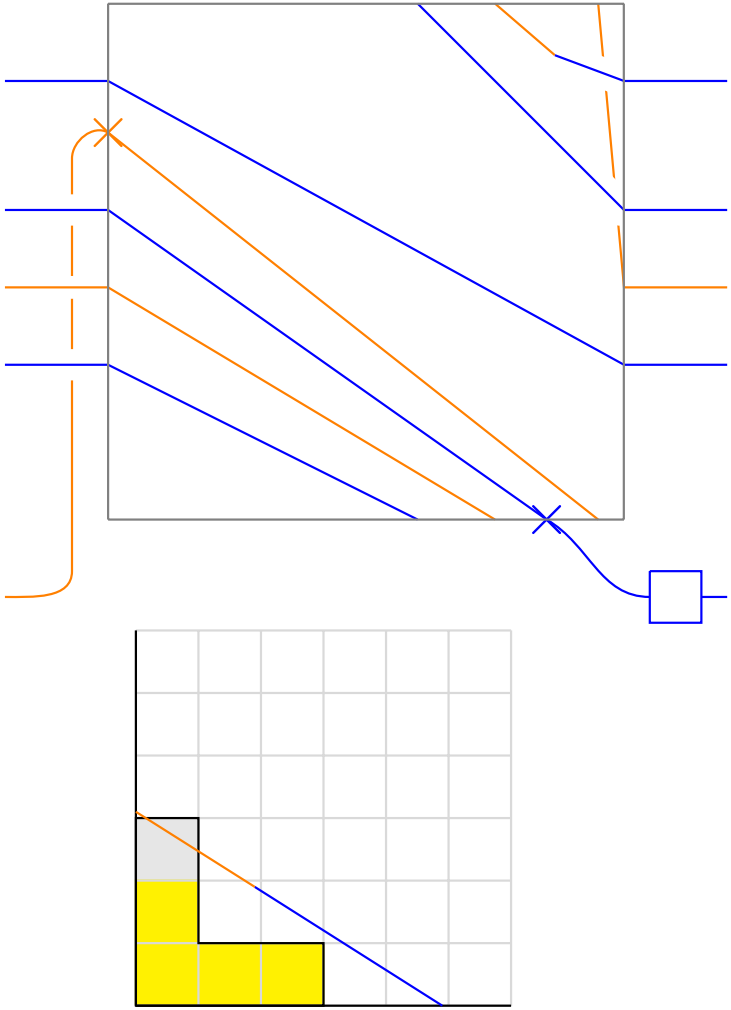

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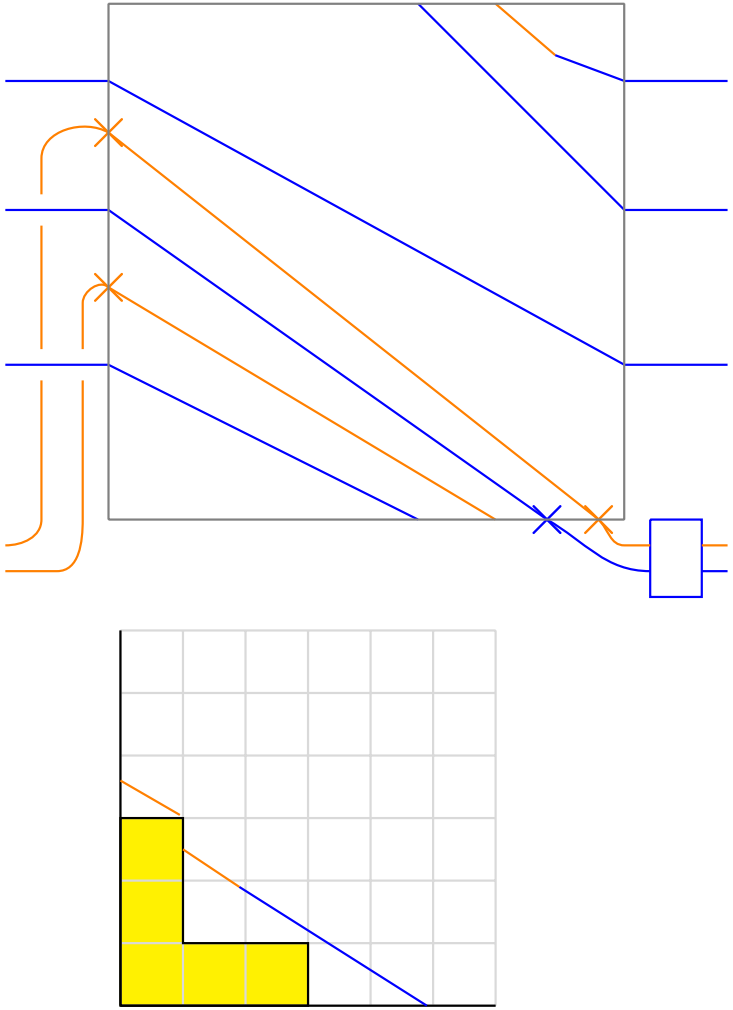

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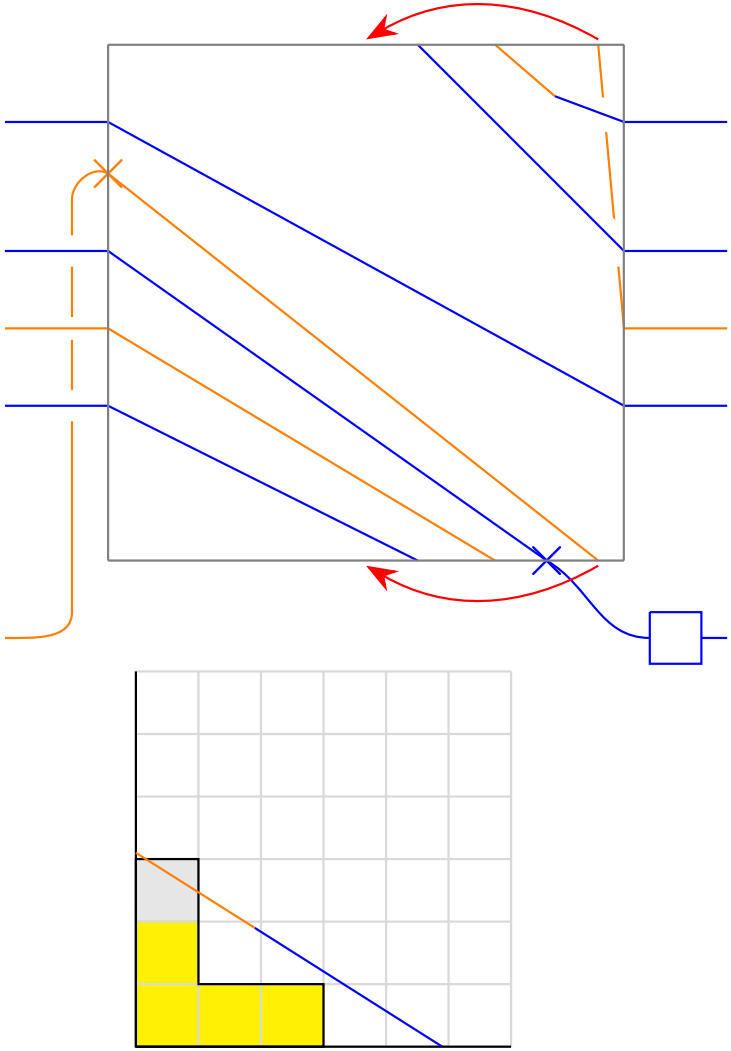

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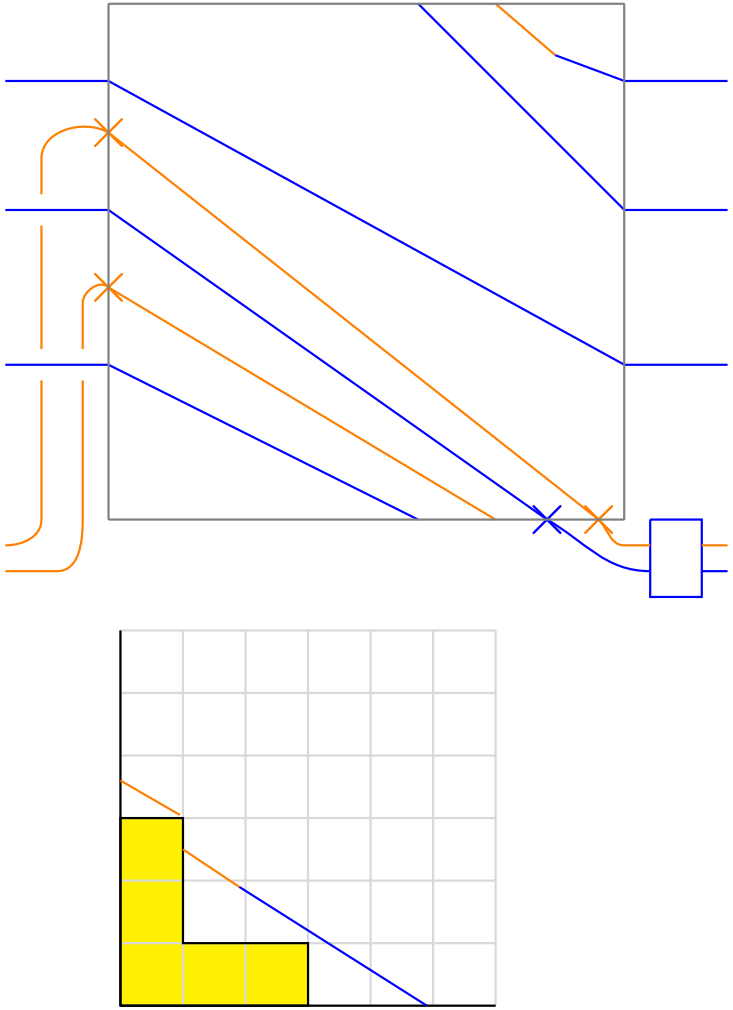

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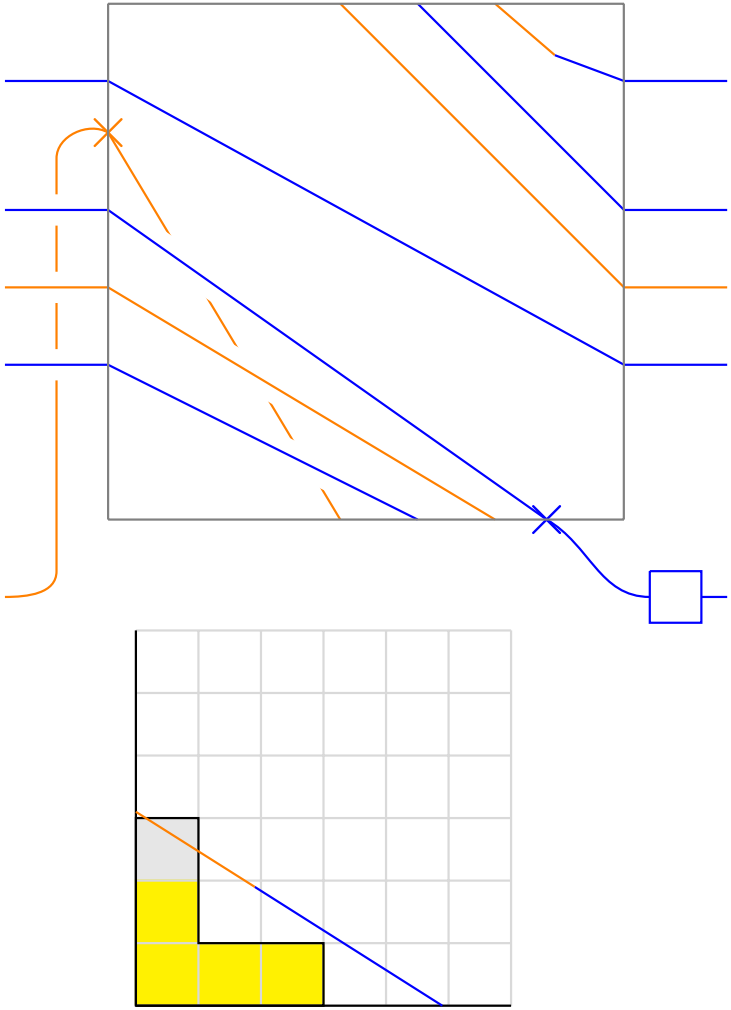

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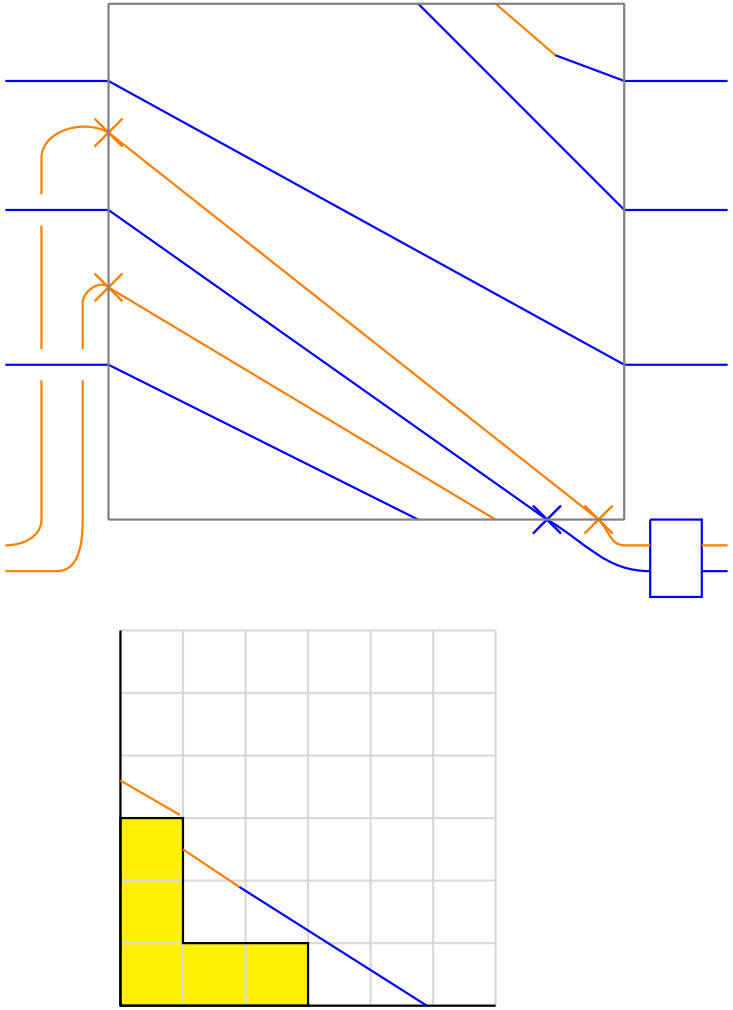

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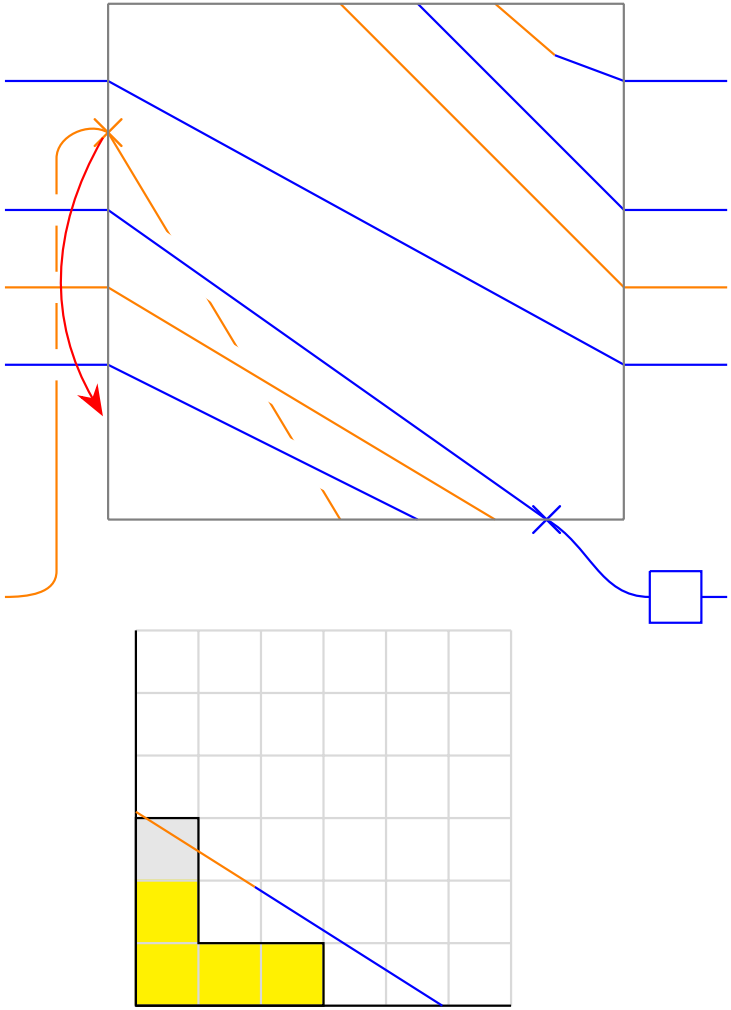

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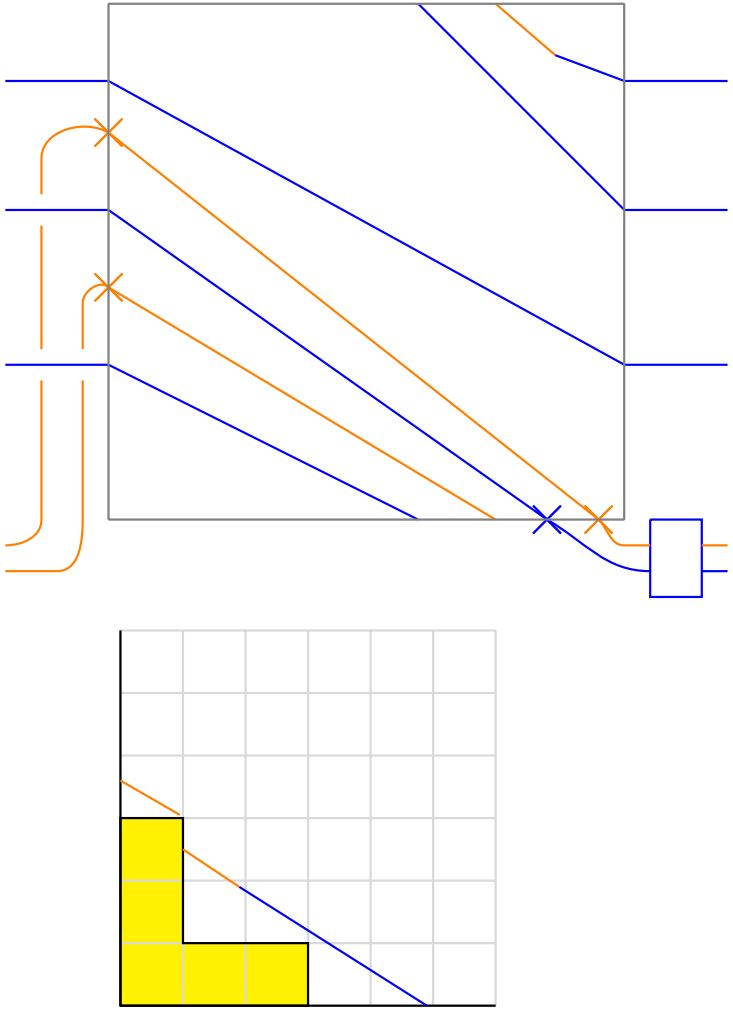

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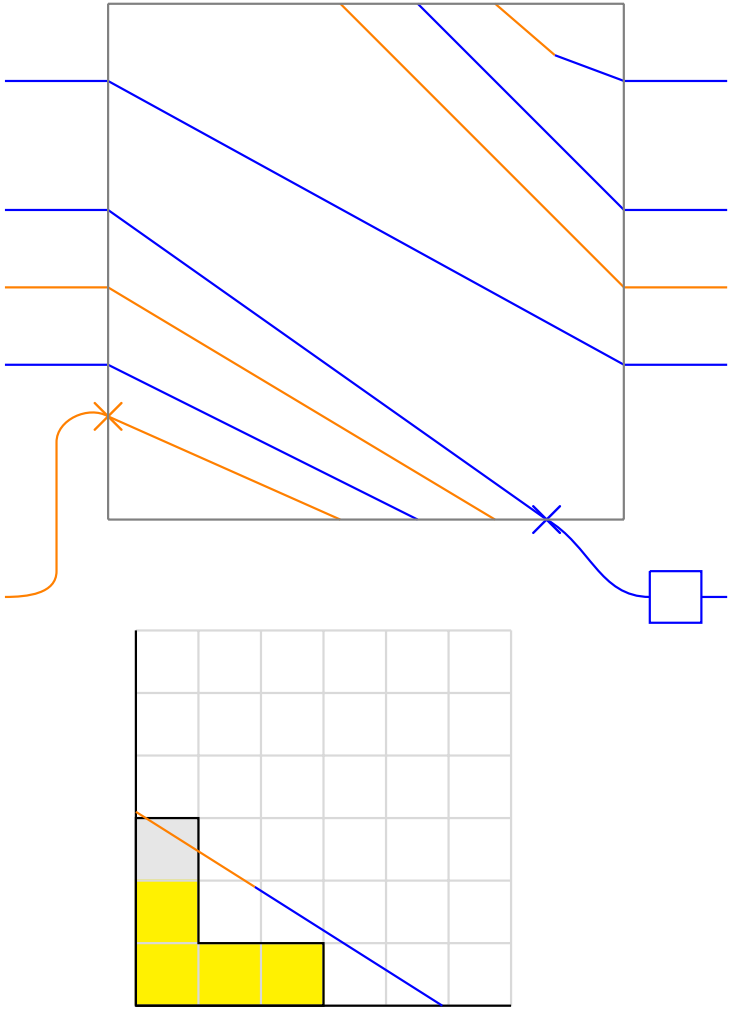

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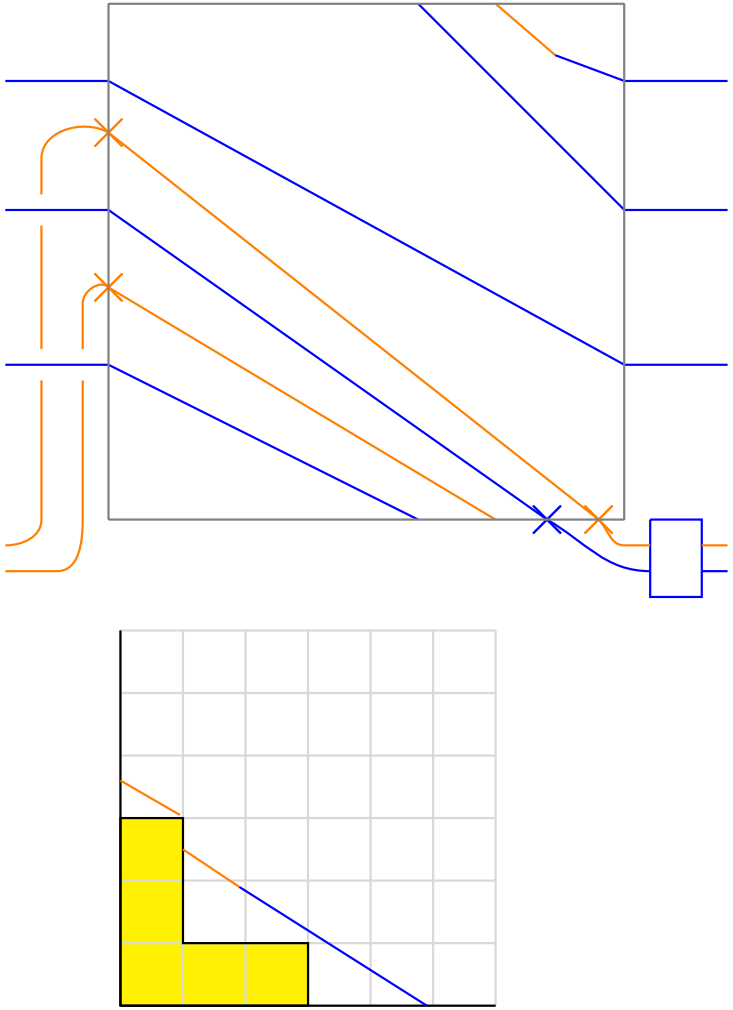
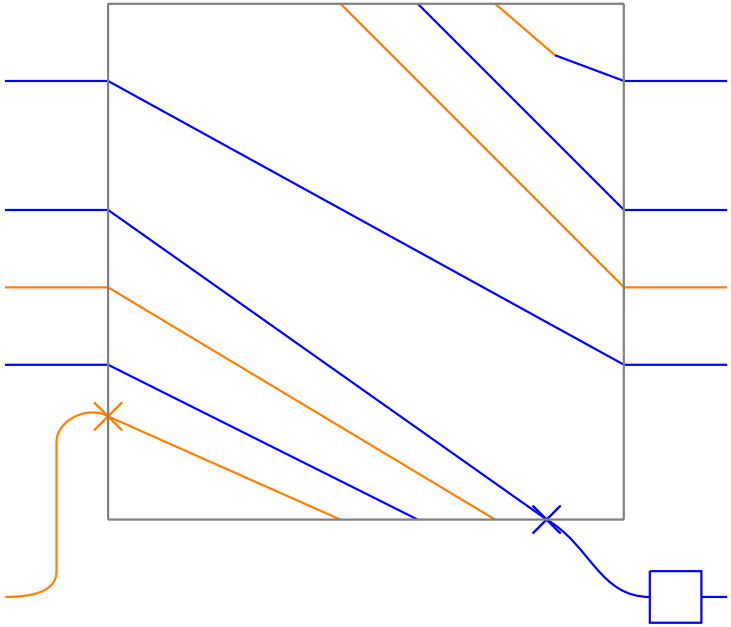
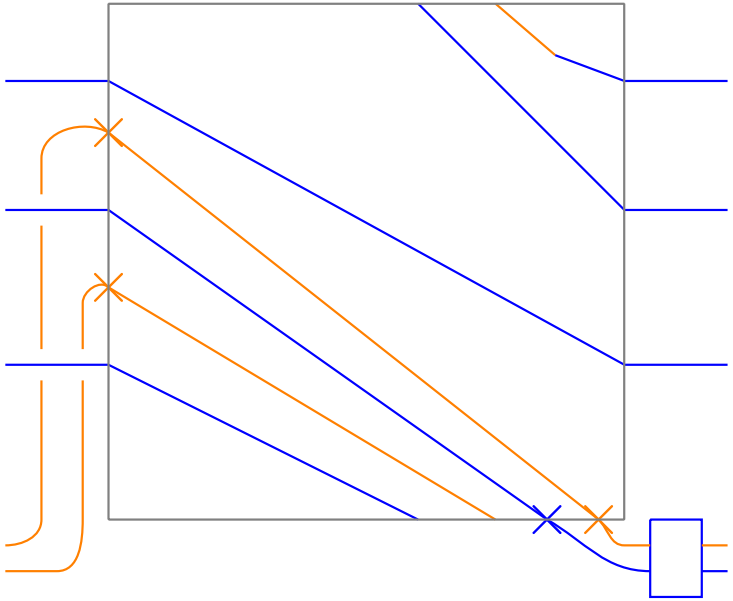
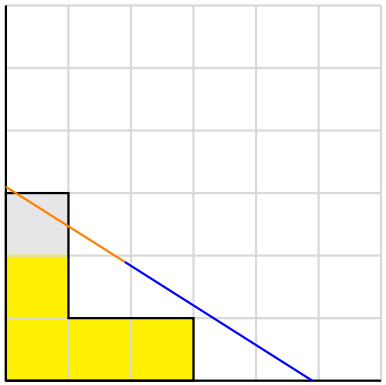
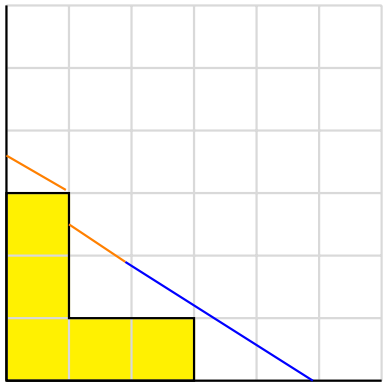
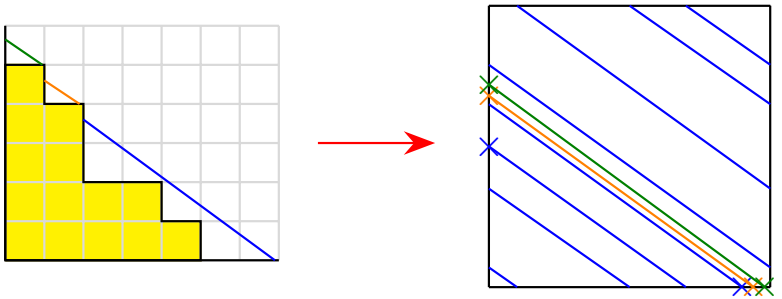


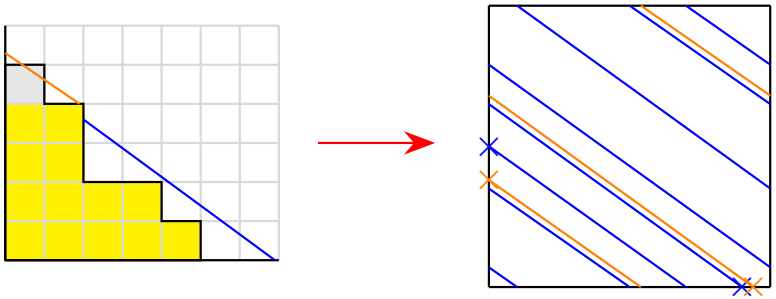
qt^{-1}  $+\ t^{-1}$ 

$$qt^{-1}R_{10000,0001} \quad + \quad t^{-1}R_{01010,0011}$$



$$t^{-3}\,T_{0010110,000111}$$



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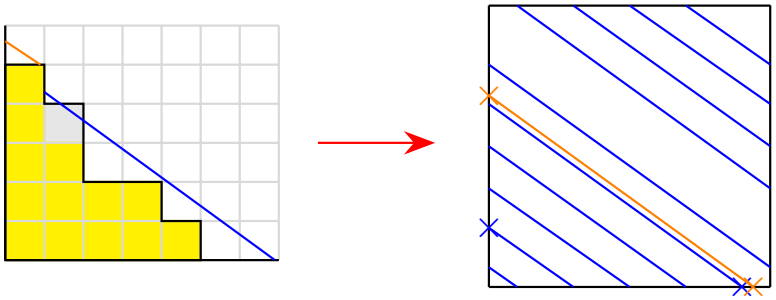
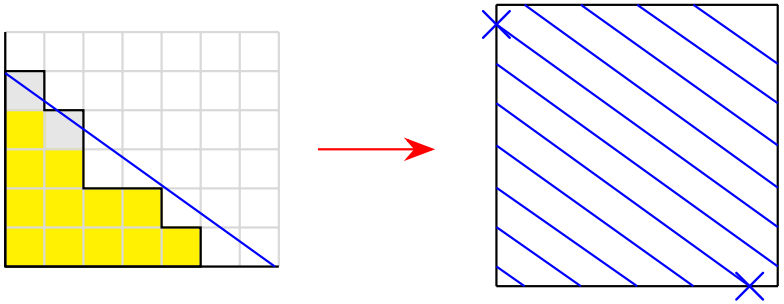
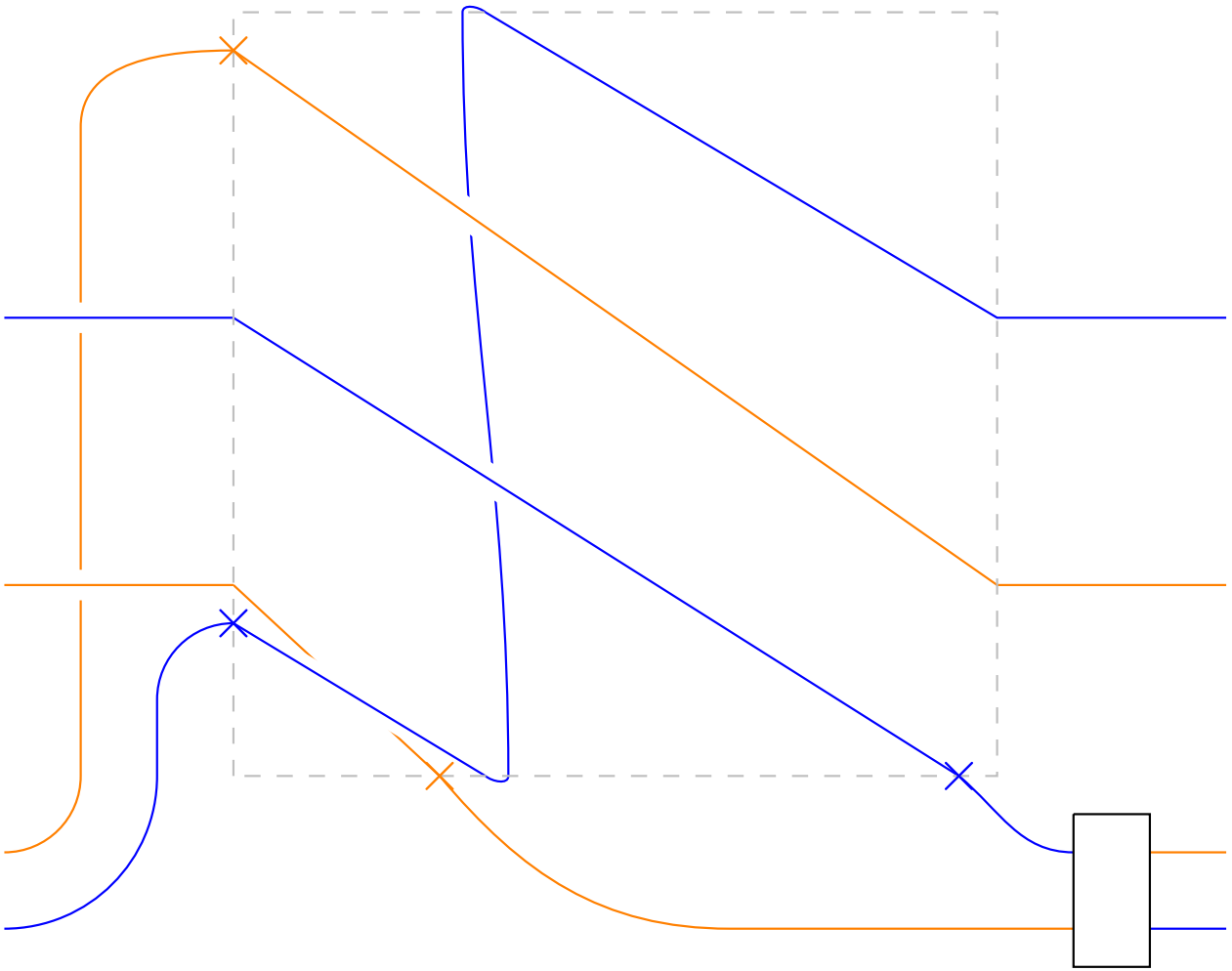


$$qt^{-2}\,T_{0100010,000011}$$



$$q^2t^{-2}\,T_{0000001,00001}$$

Troubles



Results

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Theorem (Almost)

Similar formula for two first columns.

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The Schroder polynomial $S_{\mathbf{b}}(q,t,a)$ is defined by applying $Y_{\mathbf{b}}$ to $1 \in \Lambda(q,t)$ and taking the hook components:

$$S_{\mathbf{b}}(q,t,a) := \sum_{k=0}^{n-1} a^k \langle Y_{\mathbf{b}}(1), s_{(k+1)(1)^{n-k-1}} \rangle.$$

Elliptic Hall algebra formula

One has the following explicit formula:

$$S_{\mathbf{b}}(q,t,a) = \sum_T z_1^{b_1} \dots z_n^{b_n} \prod_{i=2}^{n} \frac{1 + a/z_i}{(1 - z_i^{-1})(1 - qtz_{i-1}/z_i)} \prod_{i<j} \omega(z_i/z_j),$$

where the summation is over all standard Young tableaux of size n, $z_1, z_2 \dots, z_n$ are the q, t characters of the boxes of T labelled by $1, 2, \dots, n$ correspondingly, and the functions $\omega(x)$ is given by

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Triangular partitions follow from our work and the Shuffle Theorems.

Thank you!